

Predicting Forest Growth with Physics-Informed Deep Learning: Environmental Drivers and Interpretable Models

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Abstract

Forestry models rely on standard growth curves to show how forests develop on average, but miss why individual trees stray from that path. This dissertation combines repeated airborne LiDAR, forest records, and environmental data to track Sitka spruce top-height growth across Aberfoyle Forest, Scotland, spanning 21-years.

Linear models, mixed-effects models, and XGBoost identified factors linked to each plot's average gap from a baseline Chapman-Richards curve. A spatially varying neural model (GNNWR) tested whether these links shift across the landscape by targeting the gap between a plot's fitted height ceiling and its yield-class reference. Standard machine learning, neural networks, and physics-guided models were then compared for direct top-height prediction using compartment-level spatial cross-validation to test whether adding environmental features helps.

Canopy cover was the strongest driver of growth gaps. Management added minor signals, while broad environmental layers added little beyond local wind shelter and exposure effects. Although GNNWR slightly beat global models, canopy cover drove 80–90% of its signal, showing little extra spatial pattern from the environment. XGBoost gave the most accurate top-height predictions. PINN- k had an RMSE roughly 0.4 m higher, but it eliminated impossible predictions (5–70 m) while producing meaningful, plot-level parameters.

Local stand structure gave the clearest signals, while spatially varying models added little value. Physics guidance provided interpretable parameters rather than improving raw accuracy.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified. AI tools were used throughout this project to help plan work, research papers, brainstorm ideas, edit code for certain implementations, data analysis, plotting and submitting cluster jobs. I always checked this assistance myself, tested it, and critically evaluated its output before relying on it, so the ideas and writing in this thesis are my own. AI assistance for writing was limited to feedback and suggestions, not direct edits to the submitted text.

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Chapter 1

Introduction

1.1 Motivation and study context

Forest Digital Twins are a wider long-term goal for the forestry sector: digital models of forest systems that help monitor, forecast, and management planning. They aim to replace static, periodic checks with adaptive, data-driven planning by combining repeated field observations, simulation, and machine learning(ML), and artificial intelligence(AI) [1, 2]. This is vital as managers face climate risks like heavy storms, high winds, and severe summer heatwaves that spark wildfires and shift growth. At the same time, forestry is moving towards continuous cover management, which demands more flexible stand-level models [3].

Sitka spruce (*Picea sitchensis*) is Britain’s primary commercial species, making up approximately 50% of managed forests [4] and contributing roughly £1.1 billion annually to Scotland’s economy [5, 6]. The study focuses on Aberfoyle in Trossachs, Scotland, a Sitka spruce-dominated region with 21 years of repeated LiDAR coverage. Featuring varied topography with exposed, windy ridges to sheltered valleys, providing variation that causes different growth trajectories in different areas of the forest.

Top height is a key measure of stand development, commonly used in forestry models to estimate downstream metrics like timber volume and site productivity [4], and now can be accurately captured at scale using airborne LiDAR scans [7]. For most tree species, including Sitka spruce, height development over time is modelled by the Chapman-Richards equation [8], a sigmoid growth curve, with its parameters fixed to represent a stand’s maximum height potential and its speed of growth. A fixed parameter curve only captures the average population trend, but local trajectories can deviate from this baseline, due to factors like terrain, climate, soil, wind exposure, and

active management all critically constrain or enhance a stand's actual development [9, 10]. This raises two key questions: whether local departures from this shared curve are associated with identifiable environmental and management conditions, and whether independent attribution methods agree on which conditions matter.

Foresters need to know if a stand's assigned yield class still holds, or which local factors push growth off track. This dissertation tests whether letting these environment–growth relationships vary across space captures local growth patterns that fixed regional curves miss. Flexible machine learning models can also learn these patterns directly from data without fixed curve assumptions; we test deep neural networks (DNNs) and gradient-boosted trees (XGBoost) as a supplementary check on top-height predictions.

1.2 Research gap and contributions

While repeated airborne laser scanning has been used to model forest height growth and productivity [11], limited evidence exists on whether local environmental conditions are associated with departures from Sitka spruce's shared height-growth curve, and whether models can use these for improved predictions. Many current remote-sensing models do not account for spatial dependence between nearby-identical trees, leading to data leakage and inflated performance estimates [12, 13, 14]. Relying on black-box neural networks without explainable AI limits their use for stakeholders who require interpretable insights to support decision-making.

This dissertation covers one part of the Digital Twin idea: using attribution methods to explain local growth deviations, and checking if that explanation can help predict top height. It tests whether the effect of the environment on tree growth changes by location using a Geographically Neural Network-Weighted Regression (GNNWR) model, and compares standard neural networks against physics-informed neural networks (PINNs) for height prediction.

1.3 Objectives and research questions

Answering the three research questions required combining repeated forest surveys, LiDAR, and environmental rasters (terrain, wind, climate, and soil) into a single dataset. Together, they form a clear workflow for forest management: Q1 flags stands straying from the average growth curve and finds the environmental drivers behind it; Q2 tests whether a stand's assigned yield-class ceiling ($\Delta y_{\max}$) needs local adjustment; and

Q3 tests whether adding these site features and physics rules improves top-height predictions in the field.

Research Question & Target	Objectives & Operational Purpose
Q1: Baseline Curve Attribution <i>Target:</i> Multi-year mean residual ($\bar{r}_i$) from the shared Chapman–Richards curve	(1.1) Link persistent plot-level growth deviations to terrain, soil, wind, and management. (1.2) Test if drivers agree across Elastic Net, LMM, and XGBoost. Purpose: Flags off-track stands and tests attribution consistency across model types.
Q2: Spatial Variation in Growth <i>Target:</i> Yield-class asymptote deviation ($\Delta y_{\max}^{(i)}$)	(2.1) Test spatially varying models (GNNWR) under spatial CV to capture local patterns. (2.2) Separate genuine local environmental trends from noise and fitting artefacts. Purpose: Diagnose if a stand’s standard yield class needs revision based on local site effects.
Q3: Prediction & Physics Guidance <i>Target:</i> Direct LiDAR top height (H_{it})	(3.1) Benchmark empirical models against two physics-informed neural networks: PINN (height only) and PINN- k (height and growth rate). (3.2) Test if environmental features and physics constraints improve predictions or parameter interpretability, and compare the two PINN versions. Purpose: Finds the best operational model for future height prediction and tests if PINN- k ’s extra parameter justifies its small drop in accuracy.

Table 1.1: Overview of research questions, objectives, and wider management purpose.

Main findings Environmental and management conditions can explain some of the growth differences within Aberfoyle. Elastic Net, LMM, and XGBoost all point to similar features (Q1). GNNWR finds little that these three miss; most local patterns are just noise, or flaws in how the target was built (Q2). XGBoost is the most accurate model for predicting top height, outperforming the standard DNN and both PINN variants; the two-parameter PINN loses a little accuracy but never predicts an impossible height (Q3). Physics guidance and local spatial modelling therefore do not improve accuracy, but yield safer, more interpretable models.

Chapter 2

Related Work and Background

2.1 Shared height-growth models

Top height tracks upper-canopy development and serves as a standard metric of forest growth and site productivity. Multi-temporal airborne LiDAR surveys now allow top height to be estimated consistently across broad landscapes, though uncertainty persists in young stands, open canopies, and edge boundaries [4, 7]

The Chapman-Richards(CR) model describes height trajectories using a flexible sigmoid curve parametrised by an asymptotic ceiling ($y_{\max}$), an intrinsic growth rate (k), and a shape exponent (p). General Yield Class (GYC) groups stands by their expected timber production rate, with each class mapped to a standard height growth curve. While a single population curve offers a useful baseline, it cannot capture varying local site conditions and management histories [15, 8, 6]

2.2 Environmental controls on growth

Sitka spruce productivity responds strongly to elevation, wind exposure, soil moisture, and soil fertility: in upland northern Britain, General Yield Class falls by roughly $3.2\text{-}4.0\text{m}^3\text{ha}^{-1}\text{yr}^{-1}$ per 100 m rise in elevation, from lower temperatures and more wind exposure [16], and Irish plantations show the same pattern, peak growth on moist, fertile sites, with sharp declines on drought-prone or waterlogged soils [17].

Landscape-scale studies combine forest inventory with spatial layers, from early UK work pairing sub-compartment records with terrain and soil maps via regression [18] to more recent studies adding LiDAR, climate, and soil-type rasters. Since predictor importance shifts with spatial resolution and extent [19], and coarse soil layers often

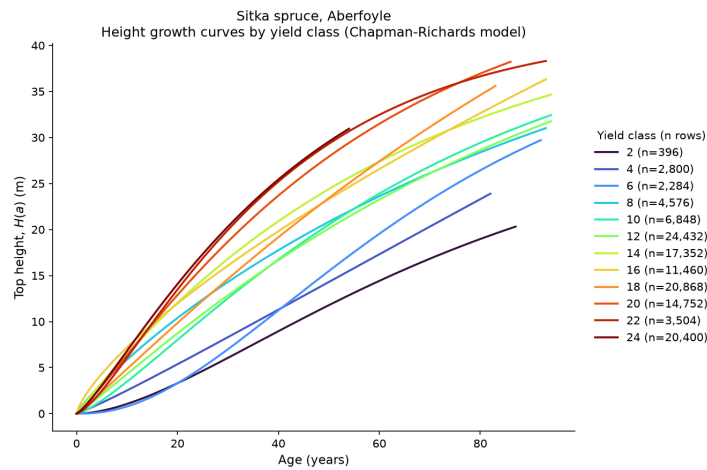


Figure 2.1: Chapman-Richards growth curves coloured by General Yield Class (GYC), work done by Forest Research

match local conditions poorly [20], differences between studies likely reflect study design rather than one universal set of drivers.

2.3 Global growth attribution

Global models apply one environment-growth rule across the entire forest rather than letting it change by location. Mixed-effects models are common in forestry for separating broad environmental trends from stand-level differences [21, 22]. However, their random effects merely soak up leftover plot variance rather than explaining its ecological causes. Elastic Net handles correlated variables well by combining penalty terms (L_1 and L_2), but it still assumes relationships are strictly linear and additive [23]. Because each method relies on different mathematical assumptions, comparing these model families is a common way to test whether an association is robust, or an artefact of one model's own structure.

Boosted-tree models have been used to map forest productivity from LiDAR and environmental data, while XGBoost-SHAP has been applied to climatic and environmental drivers of forest condition [24, 25]. When paired with tree models, SHAP has been used to identify non-linear thresholds and interactions learned by fitted tree models [26]. However, SHAP values reflect feature attribution within the model rather than direct biological causality, and attribution can be redistributed among correlated predictors [27]. Limited work compares linear, mixed-effects, and tree-based attribution on the same persistent growth-departure target.

2.4 Spatially varying attribution

Spatial autocorrelation occurs when nearby measurements are similar to one another. By contrast, spatial non-stationarity means that the relationship between the environment and tree-growth changes from one geographic area to another.

When a model leaves residual spatial autocorrelation, neighbouring areas are systematically over- or under-predicted. This may reflect missing spatial predictors, an oversimplified functional form, sampling effects, or genuine spatial non-stationarity [28].

Geographically Weighted Regression (GWR) represents non-stationarity by fitting local regressions in which nearby observations receive greater weight. Its local coefficients can be mapped to show how estimated relationships vary across space [29]. However, local multicollinearity can produce unstable or strongly correlated coefficients, making these maps difficult to interpret [30]. Multiscale GWR (MGWR) relaxes GWR's single-scale assumption by allowing each predictor to operate over a different spatial bandwidth [31].

Geographically Neural Network-Weighted Regression (GNNWR) replaces predefined spatial kernels with a neural network that learns the relationship between distance and local weighting [32]. This addresses limitations of classical GWR, whose results depend on the selected bandwidth and kernel form and whose simple weighting functions may not represent complex spatial processes. In forestry, GNNWR improved prediction and reduced residual spatial autocorrelation when relating down wood to terrain and stand characteristics [?]. However, evidence came from one forest reserve, one disturbance outcome, and five retained predictors, while the study gave limited attention to GNNWR's own interpretability and generalisability.

2.5 Physics-guided prediction

Physics-informed neural networks use governing equations as soft constraints during training [33]. Where exact physical laws are unavailable, process-guided models instead embed approximate biological knowledge. Forest applications have represented linked processes such as growth, competition, regeneration and mortality [34]. However, rigid process assumptions can bias estimates of other processes, while sparse observations may leave alternative mechanisms difficult to distinguish. Even flexible hybrid models can learn biologically implausible responses in poorly sampled conditions [34].

Training also requires empirical and process losses to be balanced. Numerical stiffness can produce unequal gradients, causing one objective to dominate optimisation [35]. Chapman-Richards guidance therefore encourages an established growth shape but does not guarantee either biological validity or improved prediction. Although process-guided networks are widely studied, integrating Chapman-Richards equations into neural networks remains rare outside specialised variable-age growth models but it has not been tested under long-term LiDAR time series in Aberfoyle.

2.6 Spatial model evaluation

Residual Moran's I is used in ecology to spot spatial patterns that average error scores can miss. It gives a simple test of whether nearby plots are consistently over- or under-predicted, indicating if a model has missed key spatial structure or yields misleading conclusions [36, 37]. In forestry, it helps compare global, GWR, and neural spatial models, where local neural models produced both lower errors and less residual clustering [38]. However, Moran's I relies on the chosen distance settings and compresses the whole map into a single number, which can hide distinct local patches. A lower score shows less error clustering under those settings, but it cannot explain why or prove that the model found a true biological relationship.

2.7 Research gap and study position

Existing studies show that the environment strongly shapes forest growth, but they typically treat growth curves, driver attribution, spatial variation, and prediction as separate problems. No previous study directly compares linear, mixed-effects, non-linear, and spatial attribution models using multi-temporal LiDAR trajectories on the same dataset with spatially independent validation. As a result, it remains unclear whether identified drivers agree across model types, whether their effects change across the landscape, and whether adding this environmental information actually improves top-height predictions.

This work addresses these gaps through three contributions:

1. **Global attribution (Q1):** Compares whether environmental associations with persistent curve departure remain consistent across Elastic Net, mixed-effects and XGBoost models.

2. **Spatial attribution (Q2):** Tests whether associations with fitted asymptotic departure vary geographically using GNNWR and spatially blocked evaluation.
3. **Prediction and process guidance (Q3):** Compares standard and neural top-height models, then tests whether environmental inputs or Chapman–Richards guidance improve accuracy or parameter interpretation.

Chapter 3

Data

This chapter describes the dataset used for model training and evaluation, outlining key feature engineering steps and data limitations. Table 3.2 summarises all predictor and target variables alongside their sources and temporal coverage. Feature selection rationale, data partitioning, and specific modelling choices are detailed in Chapter 4.

3.1 Study area and cohort coverage

The dataset covers approximately $16 \text{ km} \times 12 \text{ km}$ of Aberfoyle and contains 795,645 plot-survey records from 230,359 unique plot locations across six survey years (2002–2023). The records contain airborne laser scanning (ALS)-derived variables, including canopy cover and height percentiles, alongside stand and management fields such as planting year and known felling history (Table 3.2). Figure 3.1 shows the study-area topography, the spatial coverage of both cohorts, and an example of standard and boundary-clipped plots.

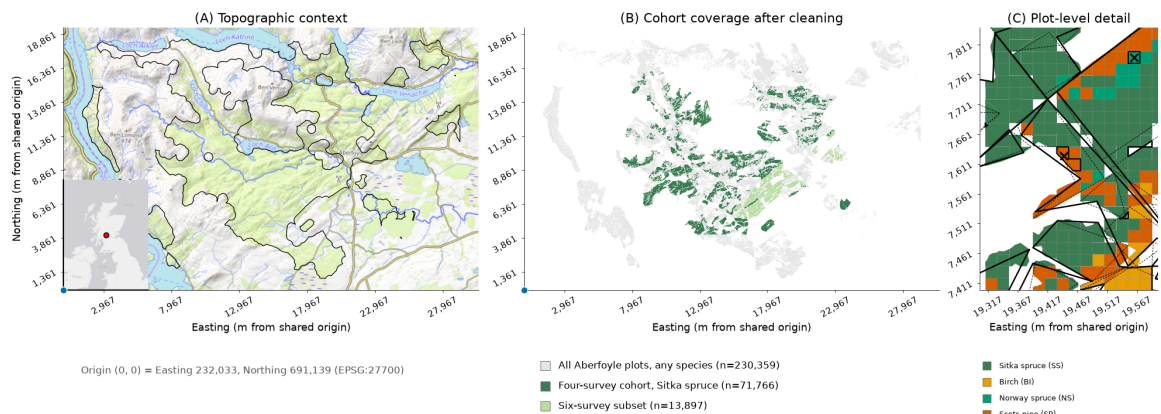


Figure 3.1: Left:Aberfoyle study area. Middle: cohort coverage, Right: plot-level detail

Because LiDAR coverage increase by 37 km² between 2006 and 2008, two nested cohorts were defined (Table 3.1): a broad-coverage *4-survey* cohort (2008, 2012, 2021, 2023; 71,766 plots across 296 compartments) and a longer-trajectory *6-survey* cohort (2002–2023; 13,897 plots across 48 compartments).

To ensure data quality, sequential filters were applied to both cohorts (Table 3.1):

- **Complete presence:** Plots must be present across all survey years in the cohort.
- **Species purity:** Pure Sitka spruce stands only.
- **Plausible records:** Planting year between 1850 and the survey year, stand age 0–200 years, and LiDAR top height 0–60 m (ecological maximum [6]).
- **Maturity cut-off:** Stand age ≥ 30 years by 2023 to exclude unreliable young-stand height measurements [7].

Stage	4-survey		6-survey	
	Plots	Removed	Plots	Removed
0. Raw dataset	230,359	,	230,359	,
1. Cohort years present	230,345	0.01%	230,359	0.00%
2. Complete survey coverage	112,782	51.04%	18,781	91.85%
3. Sitka spruce only	72,395	35.81%	13,897	26.01%
4. Valid planting year	71,766	0.87%	13,897	0.00%
5. Plausible age (0-200 years)	71,766	0.00%	13,897	0.00%
6. Plausible height (0-60 m)	71,766	0.00%	13,897	0.00%
7. Mature stands (age ≥ 30 in 2023)	58,112	19.03%	13,769	0.92%

Table 3.1: Plots left after each filter. Percentages show the share removed from the previous row, rather than from the raw dataset. Figure 3.1 shows the final spatial coverage.

3.2 Response variables and targets

All targets in this dissertation derive from the plot-level 95th percentile ALS return height (`elev_percentile_95th`), forming three distinct response variables. The multi-year mean residual from a shared growth curve ($Q1, \bar{r}_i$), the gap between a plot’s fitted ALS height ceiling and its assigned yield-class ceiling ($Q2, \Delta y_{\max}^{(i)}$), and direct LiDAR top height ($Q3, H_{it}$). Forest Research records supply planting year, management history, and yield class (Table 3.2), with any variables calculated directly from measured height removed from model inputs to prevent target data leakage. A code-level audit confirmed

that the pipeline contains no target circularity: environmental features (terrain, climate, wind, and soil) were extracted independently from external spatial layers and joined only at the modelling stage, never entering the target calculation itself. However, because historical yield-class assignments come directly from Forestry Commission records, the original field assessments that generated them could not be independently verified.

Field	Meaning or derivation	Role in this study
elev_percentile_95th	95th percentile of ALS return heights above ground, used without correction	Response variable
Top_Height95	$= \text{elev_percentile_95th} \times 1.1$; Forest Research's own corrected height estimate	Retained for reporting but not modelled
plyr, Age	Planting year; Age = survey year – plyr	Input feature; used for data cleaning
ylde	An existing Forest Research yield-class field, not calculated from this project's LiDAR-derived height response	Excluded from prediction inputs; used to construct Q2's own attribution target
whcl	Ordinal Forest Research windthrow hazard class, 0–6	Management descriptor
CanopyCover	Canopy-cover fraction from the same ALS survey and processing step as the height response	Input feature
Thin, last_thinn, next_thin_	Thinning history; Thin is binary (1 if thinned, 0 if never thinned); last_thinn/next_thin_ record last and next thinning years	Input feature
Vol95, area, g1, g3	Forest Research volume estimate, calculated from height, plot area, and canopy cover	Used for disturbance screening
GYCspec95	General Yield Class calculated directly from a plot's own measured height (Top_Height95)	Excluded (target leakage)
p1–p5	Forest Research growth curve parameters for yield class	Excluded from inputs; used with ylde to build Q2 target curve

Table 3.2: Target, stand, and management fields provided by Forest Research. GYCspec95 and p1–p5 are excluded from model inputs to prevent target data leakage. In Q2, p4 and p5 are held fixed only to define the baseline yield-class asymptote target.

3.3 Environmental data sources

Each plot was matched with external terrain, wind, climate, and soil variables (Table 3.3).

Environmental variables vary in their time resolution. Terrain, soil, climate (CHELSA, Global Wind Atlas), and TOPEX were kept static, assigning each plot a single value across all survey years. TOPEX measures topographic shelter, where higher scores mean greater shelter and lower scores mean higher exposure. By contrast, MIDAS wind metrics were calculated specifically for each interval between survey dates using

Data family	Source and resolution	Time basis	Variables used
Terrain	OS Terrain 50, 50 m [39]	Static	Elevation, slope, aspect, northness and eastness
Soil	CEH Natural Capital, 50 m [40]; SoilGrids v2.0, 250 m [41]	Static	Pedotope, drainage, texture and soil pH
Wind exposure	Global Wind Atlas, 250 m [42]	2006–2015 climate	Long-term wind speed and exposure
Observed wind	MIDAS Open weather stations [43]	Survey intervals	Mean and extreme wind conditions
Long-term climate	CHELSA v2.1, 1 km [44]	1981–2010 climate	Growing degree days and precipitation
Matched climate	HadUK-Grid, 1 km [45]	Cohort years	Temperature and frost
Spatial context	OS Open Roads and Open Rivers, vector data [46, 47]	Static	Distance to roads, rivers and forest edges

Table 3.3: External data sources used in the analysis. Values were matched to plot centres. Full source, resolution, coverage and extraction details are given in Appendix A.2.

weather-station data. HadUK-Grid temperature and frost values were averaged over the survey years within each cohort, while road, river, and edge distances were treated as static baselines. Limited missing values in environmental features were handled by removing incomplete rows for specific model sets or using missing data indicators for thinning history (Chapter 4).

Several limitations should be noted: Requiring plots to appear across every survey year ensured clean tracking over time, but reduced landscape coverage, particularly in the 6-survey cohort where high-elevation areas are under-represented. Predictor layers also vary in detail: terrain changes across individual plots, whereas regional climate maps share the same values across wide areas and do not update per survey date. Additionally, canopy cover shares the same LiDAR flight acquisitions as top height, which could introduce shared survey conditions.

Chapter 4

Methodology and Experimental Framework

4.1 Study design

Q	Unit & Target	Models	Main Evaluation
Q1	Plot; mean growth curve residual ($\bar{r}_i$)	Elastic Net, XGBoost, LMM	Spatial block CV, LMM variance decomposition, residual Moran's I
Q2	Plot; asymptote deviation from yield class ($\Delta y_{\max,i}$)	Elastic Net, XGBoost, GNNWR	5-fold spatial block CV, SHAP attribution, Moran's I , target audit
Q3	Plot–survey; top height (H_{it})	CR, Linear, RF, XGBoost, DNN, PINN, PINN- k	5-fold spatial block CV, pooled metrics, hyperparameter/ λ /feature checks, permutation importance, age/height errors

Table 4.1: Units, targets, models, and evaluation methods across research questions.

Data partitions and leakage control The 4-survey cohort was our main dataset. The smaller 6-survey cohort gave us a longer-term check. For Q1 and Q2, each plot became one summary value. For Q3, we used every plot-survey record. To stop data leakage, we split the data by forest compartment, not by plot [48]. No compartment appeared in both training and test sets. We also removed training plots within 60m of validation or test plots, to stop spatial spillover. We grouped the compartments into five equal folds. In each round, three folds trained the model, one tuned its settings, and one tested final accuracy. Across the five rounds, each compartment was tested once. All test

predictions were then pooled to get the overall scores [13].

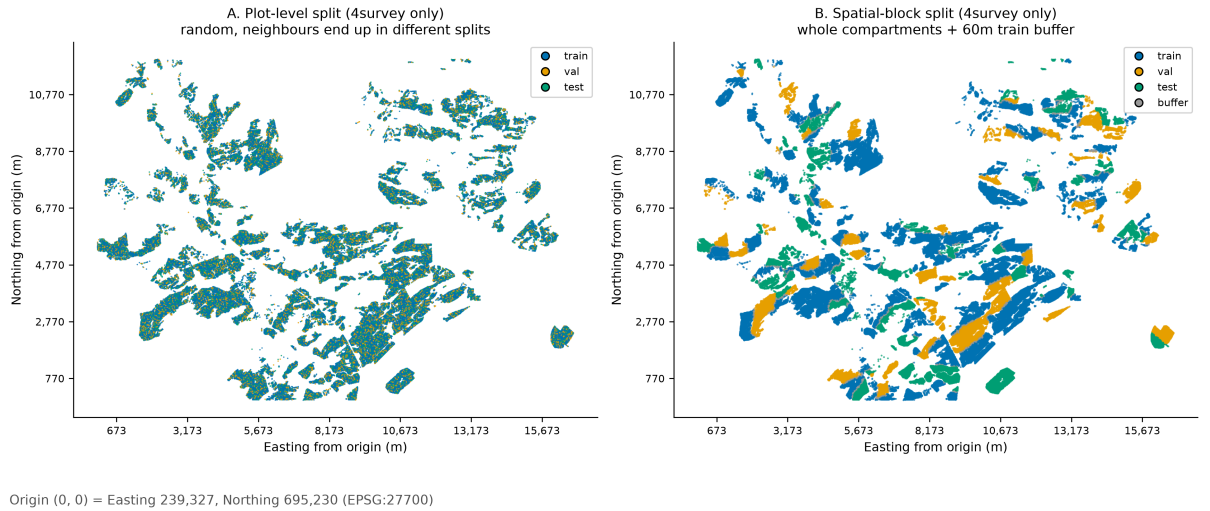


Figure 4.1: Data-splitting strategies for model evaluation: (A) Random plot-level split: shown for comparison only, as spatial proximity between adjacent training and test plots caused severe data leakage and inflated accuracy. (B) Compartment-level spatial-block split with a 60 m buffer: the primary strategy used throughout this dissertation to ensure complete spatial independence between training, validation, and test sets.

Continuous variables were normalised using training-set statistics and applied to validation and test folds. Stand age was scaled separately so PINN outputs could be directly converted back to metres per year. Soil classes were one-hot encoded, while administrative IDs (compartment and block) were excluded from all feature sets to prevent data leakage.

4.2 Feature-set construction

Four feature sets were constructed to test whether adding environmental data improves model performance. Because feature relevance depends on the specific prediction task, sets were ranked and selected separately for Q1, Q2, and Q3. Variables with extensive missing values and exact duplicates were removed first. For highly correlated pairs (Spearman $|\rho| \geq 0.95$), we retained only the directly measured variable. The remaining candidates were ranked using their average rank across three metrics evaluated on a single spatial split: Spearman $|\rho|$, XGBoost permutation importance, and XGBoost drop-column ablation. To prevent multicollinearity, we iteratively removed variables with a variance inflation factor (VIF) greater than 5 until all remaining features met

the threshold [49]. Baseline stand variables were kept in every feature set but still checked for multicollinearity in the VIF calculations. This screening removed 430 four-survey plots, primarily due to gaps in SoilGrids coverage; the six-survey cohort had complete environmental coverage. Full missingness summaries and exact feature lists are provided in Appendix B.

Set	Construction
Set 1	Stand and management baseline only.
Set 2	Set 1 plus the top 10 ranked environmental candidates (candidates failing VIF screening were replaced by the next-ranked variable).
Set 3	Set 1 plus the top 50% of terrain and wind features, followed by VIF screening.
Set 4	Set 3 candidates plus the top 50% of climate, soil, and spatial-context features, followed by VIF screening across the pooled set.

Table 4.2: Construction of environmental feature sets: Feature ranking was performed independently for Q1, Q2, and Q3

Because VIF screening was rerun whenever the candidate pool expanded, the final sets are not strictly nested. For example, elevation was selected in Sets 2 and 3 but dropped from Set 4 due to collinearity with broader terrain and climate predictors. This reflects statistical redundancy within the larger pool rather than an absence of ecological influence.

4.3 Q1: Global departure attribution

Q1 evaluates which forest stands consistently outperform or lag behind the regional (CR) growth curve once age is accounted for, and identifies the environmental drivers behind these differences. Repeated survey heights are condensed into a single mean residual per plot, and Elastic Net, XGBoost, and a linear mixed-effects model are trained to predict this deviation. We use the 4-survey cohort because the 6-survey cohort contains only 47 compartments, which is insufficient for robust spatial-block cross-validation.

What Q1's target means in plain terms Let H_{it} and a_{it} denote the observed top height and stand age for plot i at survey t , where T_i is the total number of surveys for

plot i . Using the regional baseline height $H(a_{it})$ from Equation (4.10), the survey-level residual r_{it} and plot-level mean departure $\bar{r}_i$ are defined as:

$$r_{it} = H_{it} - H(a_{it}), \quad (4.1)$$

$$\bar{r}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} r_{it} \quad (4.2)$$

A positive $\bar{r}_i$ indicates that a plot tracks above the regional baseline on average, while a negative value indicates under-performance. Because $\bar{r}_i$ reflects average vertical elevation relative to the curve rather than instantaneous growth rate, a plot's residual sign may vary across individual survey years. Stand age is excluded from the feature set since the baseline curve already removes the age trend.

Models Both Elastic Net and XGBoost are evaluated using 5-fold spatial-block cross-validation. Elastic Net tunes its regularisation penalty and ℓ_1 -ratio via inner 5-fold cross-validation (`ElasticNetCV`). XGBoost uses fixed hyperparameters (max depth 4, learning rate 0.04, up to 500 trees with early stopping), adopted after initial validation checks rather than exhaustive re-tuning on this specific target (the direct top-height XGBoost in Q3 uses its own targeted grid search). Model performance is reported using out-of-fold pooled metrics (R^2 , RMSE, and MAE).

To account for plots grouped within compartments, a linear mixed-effects model is fit on each fold's training partition:

$$\bar{r}_i = \beta_0 + \beta^\top \mathbf{x}_i + u_{c[i]} + \epsilon_i, \quad u_c \sim \mathcal{N}(0, \sigma_u^2). \quad (4.3)$$

Here, $\beta^\top \mathbf{x}_i$ is the fixed-effects term describing broad relationships across all plots (such as slope or TOPEX), and $u_{c[i]}$ is a random intercept for plot i 's compartment: each training compartment gets its own value, drawn from a shared distribution with variance σ_u^2 , capturing grouping the model cannot otherwise explain (without saying why it exists). This model is used only for its fitted coefficients and the variance-explained comparison below, not to predict height for unseen compartments.

Parameters (β, σ_u^2) are fitted with restricted maximum likelihood (REML), which avoids underestimating variance. Comparing a null model (compartment intercept only) against the full model (with fixed effects added) instead uses ordinary maximum likelihood, since REML fits cannot be compared across different fixed-effect structures; variance explained is the share of the null model's compartment-level variance that shrinks once fixed effects are added. The model is refit per fold, so coefficients are

reported as mean $\pm$ SD across folds, showing fold-to-fold stability, unlike the pooled Elastic Net/XGBoost accuracy metrics above.

Sensitivity checks Canopy cover and top height come from the same LiDAR flights, so a strong canopy-cover effect could reflect measurement overlap rather than real stand structure. Three steps test this: refit Set 4 without canopy cover and check the change in R^2 ; check the direct correlation between canopy cover and each target; and recompute SHAP values without canopy cover to see if other predictors take over its role.

Terrain coefficients for slope and TOPEX also sometimes differ between Elastic Net and the mixed-effects model, which could mean local compartment effects that global models miss. A further check measures how much slope and TOPEX variance sits between compartments versus within them (the intraclass correlation coefficient, ICC).

Elastic Net coefficient signs, mixed-model fixed-effect signs, and XGBoost mean |SHAP| rankings are then compared. This shows which variables matter, and in which direction when sign is available. The raw sizes are not directly comparable because the models differ, and mean |SHAP| has no sign. Agreement across all three makes an association less likely to be a quirk of one model, but it does not show causality. Residual Moran's I checks for remaining spatial clustering (Section 4.6).

4.4 Q2: Spatial asymptote attribution

Q2 builds on Q1 by asking not just which variables matter, but whether their effect changes by place. Global models assume one environment-growth relationship for the whole study area; Q2 tests whether letting this relationship vary by location explains more and reduces leftover spatial clustering [32]. Each plot gets a prediction based on its location, on the idea that nearby plots behave more alike than distant ones, a neural network learns these location-based weights directly from coordinates, with no fixed neighbourhood size chosen in advance. GNNWR still assumes each local relationship is a straight line, so a changing local coefficient might just mean the straight-line fit is poor there, not a real difference in growth drivers by place. For that reason, both accuracy and the local coefficients are considered together.

What Q2's target means in plain terms Q2 asks whether a plot grows towards a higher or lower final height than its official yield class predicts. p_1 – p_5 are Forest

Research's published numbers that convert a yield-class rating into a full growth curve for a species and region. This project did not derive them, just reused them. p_1 – p_3 set the curve's asymptote (ceiling); p_4 – p_5 set its shape (rate and curvature), two parts of one sigmoid. p_4 and p_5 are fixed at their published values for the matching Forest Research curve; only the height ceiling is allowed to differ from plot to plot. Earlier work has represented environmental variation through the Chapman–Richards asymptote in the same way [50]. This supports it as a simple and workable choice. For plot i at survey t :

$$g_{it} = (1 - e^{-p_{4,i}a_{it}})^{p_{5,i}}, \quad (4.4)$$

$$\hat{y}_{\max,i} = \frac{\sum_t H_{it} g_{it}}{\sum_t g_{it}^2}, \quad (4.5)$$

$$y_{\max,i}^{\text{yldc}} = \text{yldc}_i p_{2,i} + p_{1,i} + 2p_{3,i}, \quad (4.6)$$

$$\Delta y_{\max,i} = \hat{y}_{\max,i} - y_{\max,i}^{\text{yldc}}. \quad (4.7)$$

With p_4 and p_5 fixed, Equation (4.5) gives a least-squares estimate for $\hat{y}_{\max,i}$. This avoids trying to estimate three correlated curve parameters from only 4–6 observations per plot. $\hat{y}_{\max,i}$ is the ceiling fitted from plot i 's own observed trajectory, $y_{\max,i}^{\text{yldc}}$ (Equation (4.6)) is the ceiling implied by its assigned yield class, using the same fixed rate and shape. Their difference, $\Delta y_{\max,i}$ (Equation (4.7)), is positive when the fitted ceiling is higher than expected and negative when it is lower. This target is estimated, not directly observed.

Following the Q1 setup, Elastic Net and XGBoost are trained on $\Delta y_{\max}$ using Sets 2–4, with the same tuning approach as Q1. Elastic Net tests a global linear relationship. XGBoost tests a global non-linear relationship. GNNWR tests a spatially varying but locally linear relationship. Spatial variation is modelled using GNNWR via its Python library [32, 51]:

$$\widehat{\Delta y}_{\max,i} = \beta_0(s_i) + \sum_j \beta_j(s_i) x_{ij}. \quad (4.8)$$

For each location s_i , GNNWR uses a neural network to learn spatial weights from distances to training plots. These weights determine the local coefficients $\beta_j(s_i)$ around a global linear baseline. Each training fold uses only that fold's own training partition. No hyperparameter search was performed for GNNWR. Every run uses the library's default architecture and training settings.

Computational adjustments and feature evaluation

Headline GNNWR runs used full reference sets (earlier GPU-memory caps are retained only for sensitivity checks). To bypass memory limits, hat-matrix diagnostics were

disabled; consequently, GNNWR AIC uses raw feature counts rather than effective degrees of freedom, and F -tests are omitted (predictive metrics, Elastic Net, and XGBoost are unaffected). Feature importance is evaluated via XGBoost gain, test-fold SHAP values [26], and GNNWR local coefficient distributions and maps. Feature-group contributions to R^2 use permutation importance (shuffling columns without refitting), while residual clustering impacts are measured by removing groups from Set 4, refitting, and comparing Moran's I on the four-survey cohort. Finally, canopy cover dependency is isolated by refitting all three models using canopy cover exclusively and with it removed.

Target and outlier audit Plots affected by harvesting or clear measurement errors are screened using two rules across consecutive surveys: clearfelling requires drops in top height $\geq 25\%$, canopy cover $\geq 70\%$, and volume $\geq 70\%$, while measurement error is flagged when top height drops $\geq 25\%$ alongside canopy cover drops $\leq 10\%$ and stable volume; unclassified height drops are retained. Extreme fitted height ceilings ($\hat{y}_{\max,i}$) are flagged using Tukey's $1.5 \times \text{IQR}$ fences [52],

$$[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}], \quad (4.9)$$

These bounds are calculated per cohort, Flagged plots remain in main models but are evaluated separately in a holdout check. To determine if extreme errors link to fitting artefacts, disturbance, or environment, we compare overlap among the top 1% and 5% residual plots across Elastic Net, XGBoost, and GNNWR, examining their spatial spread, edge distance, wind shelter, and local coefficients. Results are detailed later.

4.5 Q3: Top-height prediction

Q3 compares models for predicting top height: a baseline growth curve, standard tabular models, and neural networks, including physics-informed ones. It then tests whether environmental variables or Chapman–Richards(CR) guidance improve accuracy or the interpretability of the fitted parameters. Using five spatial folds on Set 3, the other feature sets test whether adding environmental data helps.

CR baseline

$$H(a) = y_{\max} \left(1 - e^{-ka}\right)^p, \quad (4.10)$$

where $H(a)$ is height at age a , $y_{\max}$ is the height ceiling, k is the growth rate, and p controls curve shape. All three parameters are fit freely by non-linear least squares on each training fold and then frozen, keeping test folds fully independent.

Conventional models We test three baselines: linear regression, random forest, and XGBoost. For Q3, XGBoost was tuned across 27 parameter settings over five folds using validation R^2 .

DNN and physics-informed models All neural networks: three-layer trunk, 128 units per layer, Leaky-ReLU, Adam optimiser (learning rate 10^{-4} , weight decay 10^{-5} , batch size 256, early stopping), defaults, checked against alternatives (Sensitivity checks and diagnostics, below). Standard DNN: minimises mean squared error ($\mathcal{L}_{\text{data}} = \frac{1}{N} \sum (\hat{H}_i - H_i)^2$). Conditioned DNN: environmental variables feed the input directly. PINN, PINN- k : environmental variables feed small sub-networks instead, which adjust each plot's own growth-curve parameters. Plain PINN adjusts only the height ceiling ($y_{\max}$); growth rate (k) and shape (p) stay one shared value for every plot. PINN- k adjusts both $y_{\max}$ and k ; p stays shared for both variants. These plot-level parameters adjust the predictions and loss terms together, balancing fit to the data with the baseline growth curve.

Equations (4.11)–(4.14): the physics-informed loss.

$$H'(a) = y_{\max} p k e^{-ka} \left(1 - e^{-ka}\right)^{p-1}, \quad (4.11)$$

$$\mathcal{L}_{\text{phys}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{\partial \hat{H}_i}{\partial a_i} - H'(a_i) \right)^2, \quad (4.12)$$

$$\mathcal{L}_{\text{traj}} = \frac{1}{M} \sum_{j=1}^M \left(\frac{\hat{H}_{j,\text{later}} - \hat{H}_{j,\text{earlier}}}{\Delta a_j} - H'(\bar{a}_j) \right)^2, \quad (4.13)$$

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{traj}} \mathcal{L}_{\text{traj}} + 10^{-5} \sum_{\theta} |\theta|. \quad (4.14)$$

Equation (4.11): theoretical growth rate, shown with the shared global parameters; conditioned models substitute each plot's own adjusted parameters instead, $y_{\max}^{(i)}$ for both PINN variants, plus $k^{(i)}$ for PINN- k . Equation (4.12): gap between this rate and the network's own derivative ($\partial \hat{H}_i / \partial a_i$). Equation (4.13): gap between consecutive-survey growth and this rate, at midpoint age $\bar{a}_j$. Equation (4.14): data loss plus weighted physics and trajectory terms, plus an L1 penalty. Defaults: $\lambda_{\text{phys}} = \lambda_{\text{traj}} = 1$; 0 turns the physics terms off.

Sub-networks adjust plot parameters:

$$y_{\max}^{(i)} = y_{\max} + f_y(\mathbf{x}_i), \quad k^{(i)} = k \exp(f_k(\mathbf{x}_i)). \quad (4.15)$$

Plain PINN: $y_{\max}^{(i)}$ line only, $k^{(i)} = k$ fixed. PINN- k : both lines. Each sub-network in use (f_y alone for PINN; f_y, f_k for PINN- k): one hidden layer, 16 units, turns terrain features into one number. Trunk and sub-networks train together: one Adam optimiser, one learning rate, one weight decay, no separate tuning step per part. Sub-network outputs feed the prediction directly (Equation (4.16)), so ordinary training already shapes them; λ_{phys} adds a further pull towards matching the theoretical growth rate.

The final prediction: the CR curve at the plot's adjusted parameters, plus a residual from the trunk:

$$\hat{H}_i = y_{\max}^{(i)} \left(1 - e^{-k^{(i)} a_i}\right)^P + \text{trunk residual}. \quad (4.16)$$

$y_{\max}^{(i)}$ shapes the prediction directly for both variants; $k^{(i)}$ does too, for PINN- k .

Sensitivity checks and diagnostics

Beyond 5-fold spatial evaluation, six targeted diagnostics test the robustness of the PINN and PINN- k models (single run per fold/condition):

- Training and physics settings: Evaluated alternative learning rates, batch sizes (512 vs. 256), and eight physics weights ($\lambda_{\text{phys}} = \lambda_{\text{traj}} \in [0, 3.0]$) across all models.
- Feature-set sensitivity: Tested models without environmental inputs and across Sets 2, 3, and 4 to assess input necessity and feature selection sensitivity.
- Terrain permutation importance: Shuffled individual terrain/wind inputs (10 repeats) to quantify their impact on personalised $y_{\max}$ and k predictions.
- Age/height stratification: Compared MAE across age and height bands to evaluate error distributions at juvenile and mature extremes.
- Flagged compartment tracking: Monitored a large-departure compartment across all five folds (in-sample vs. out-of-sample) to separate true site signals from fold-specific overfitting.
- Yield-class alignment: Correlated predicted parameter deviations against withheld Forest Research yield classes across the full test set to verify biological realism.

4.6 Evaluation and robustness

Model accuracy is evaluated using RMSE, MAE, R^2 , and Bias. Here $e_i = y_i - \hat{y}_i$, where y_i is the relevant target: $\bar{r}_i$ for Q1, $\Delta y_{\max,i}$ for Q2, and top height for Q3.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum e_i^2}, \quad \text{MAE} = \frac{1}{N} \sum |e_i|, \quad R^2 = 1 - \frac{\sum e_i^2}{\sum (y_i - \bar{y})^2}, \quad \text{Bias} = \frac{1}{N} \sum e_i. \quad (4.17)$$

Positive bias indicates under-prediction. We report fold-level means ($\pm$ SD) or pooled metrics across all out-of-fold predictions. For the 4-survey cohort, R^2 95% confidence intervals are calculated using 1,000 compartment bootstrap resamples.

Residual spatial autocorrelation is measured using global Moran's I :

$$I = \frac{N \sum_i \sum_j w_{ij} (e_i - \bar{e})(e_j - \bar{e})}{W \sum_i (e_i - \bar{e})^2}. \quad (4.18)$$

Spatial weights (w_{ij}) use binary, row-standardised distance bands matched to each model's own semivariogram range (up to 5,000 m; 999 permutations). When needed, rows are subsampled to 5,000 to keep this fast to compute. Moran's I is left out if semivariance (the difference between pairs of points at increasing distances) is flat. It is marked provisional if semivariance keeps rising past 5,000 m. Distances are reported alongside every value. Because each model's band is fitted to its own residual surface, Moran's I values are comparable in direction across models, but not as exact like-for-like magnitudes.

Seed and split sensitivity checks were applied selectively based on computational limits. For Elastic Net and XGBoost, models were run across eight spatial fold splits for both cohorts to test sensitivity to boundary definitions. GNNWR was tested across three splits on the 6-survey cohort (omitting random training seeds), while DNN, PINN, and PINN- k used a single seed per fold, relying instead on architectural and physics-weight tests for stability evaluation. For PINN- k , parameter disentanglement between final height ($y_{\max}$) and growth rate (k) was assessed via correlation across 2,000 compartment bootstraps, reported alongside R^2 for the main setup and physics-weight tests.

Chapter 5

Results and Discussion

5.1 Q1: Global departure attribution

Using the 4-survey cohort, canopy cover and thinning history dominate the attribution in every feature set and every method tested (Table 5.1).

Set	n plots	EN R^2 [95% CI]	XGB R^2 [95% CI]	LMM spatial var. explained
Set1 (baseline only)	71,766	0.220 [0.180, 0.252]	0.222 [0.183, 0.252]	0.016±0.078
Set2	71,766	0.359 [0.310, 0.396]	0.416 [0.355, 0.461]	0.204±0.066
Set3	71,727	0.325 [0.272, 0.365]	0.375 [0.321, 0.414]	0.049±0.048
Set4	71,330	0.358 [0.305, 0.398]	0.395 [0.338, 0.438]	0.175±0.064

Table 5.1: Attribution of the persistent mean curve residual (Equation 4.2), 4-survey cohort, spatial-block split. EN = Elastic Net; XGB = XGBoost; LMM = linear mixed-effects model. Set1 = canopy cover + thinning only (baseline); Set2 = top-10 ranked environmental predictors; Set3 = terrain/wind only; Set4 = full environmental set (terrain + climate + soil). R^2 95% CIs from 1,000 compartment bootstrap resamples. LMM spatial variance explained = share of variance attributable to the compartment random intercept on top of that set's fixed effects (mean $\pm$ SD across folds, not a bootstrap CI).

These R^2 scores are modest: even the best model (Set 2, XGBoost, $R^2 = 0.416$) leaves over half the growth variance unexplained. The goal is to uncover key environmental drivers rather than make exact plot predictions. Note that R^2 is not directly comparable across Q1, Q2, and Q3 because each uses a different target with its own variance. Figure 5.1 illustrates these persistent curve departures in the observed data across six example plots spanning the residual range.

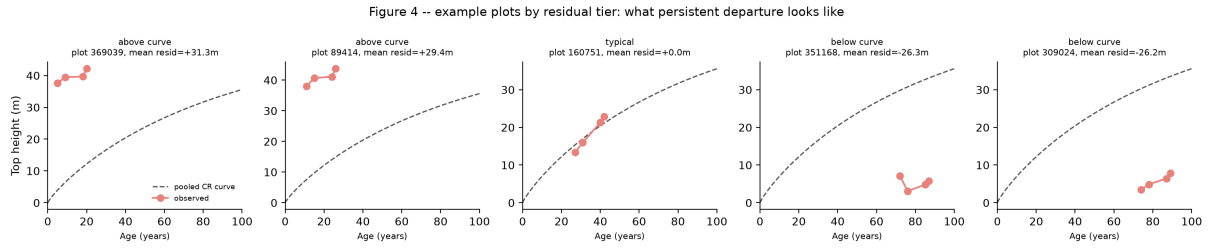


Figure 5.1: Example plots by residual tier. Observed top height vs. age (solid) against the pooled Chapman–Richards curve (dashed), for five 4-survey plots: two near the 90th percentile of $\bar{r}_i$ (above curve), one near zero (typical), two near the 10th percentile (below curve).

5.1.1 Discussion

Canopy cover and thinning history dominate attribution across all models and feature sets.

Canopy cover has the largest positive Elastic Net coefficient (Set 4: $+1.943 \pm 0.034$) and the top XGBoost importance ($|\text{SHAP}| = 1.375\text{--}1.442$). Since both canopy cover and the target come from the same LiDAR survey, we ran three checks to see if it is just a proxy for the target itself (Figure 5.2). Only moderate correlation ($r = 0.35\text{--}0.47$) rules out a simple duplicate. Removing canopy cover drops R^2 by roughly a third (Elastic Net: $0.350 \rightarrow 0.231$; XGBoost: $0.388 \rightarrow 0.249$; Figure 5.2, left), showing it adds real, unique information. No other variable steps in to fill the gap either; $\text{dist}_{\text{to_road}}$'s importance barely moves ($|\text{SHAP}| = 1.003 \rightarrow 0.869$; Figure 5.2, right). So canopy cover and thinning history are treated as a structural baseline, and the real attribution question is what else adds on top of them.

Most Set 4 predictors agree across models, but six exhibit sharp divergences

anchored by *topex* (Figure 5.3; Appendix C). While *slope_degrees* and rainfall remain consistently positive across LMM and Elastic Net (slope: $+0.86\text{--}+1.23$), *topex* splits sharply (LMM $+1.32$ vs. Elastic Net -0.18 to $+0.06$). This is not driven by multicollinearity (unstable predictors have lower VIFs, 2.1–3.4, than stable rainfall's 3.91) or climate-data timing (the static rainfall normal is stable, while the survey-matched *tas_mean* is unstable). All unstable predictors show high between-compartment clustering (ICC = 0.74–0.89), yet stable rainfall's ICC is even higher (0.97), so clustering alone cannot explain the divergence, the real mechanism follows below.

Within-compartment variation separates genuine local signals from spatial artefacts.

Because both models assume linear relationships, their divergence stems from spatial handling: LMM uses random compartment intercepts, whereas Elastic Net

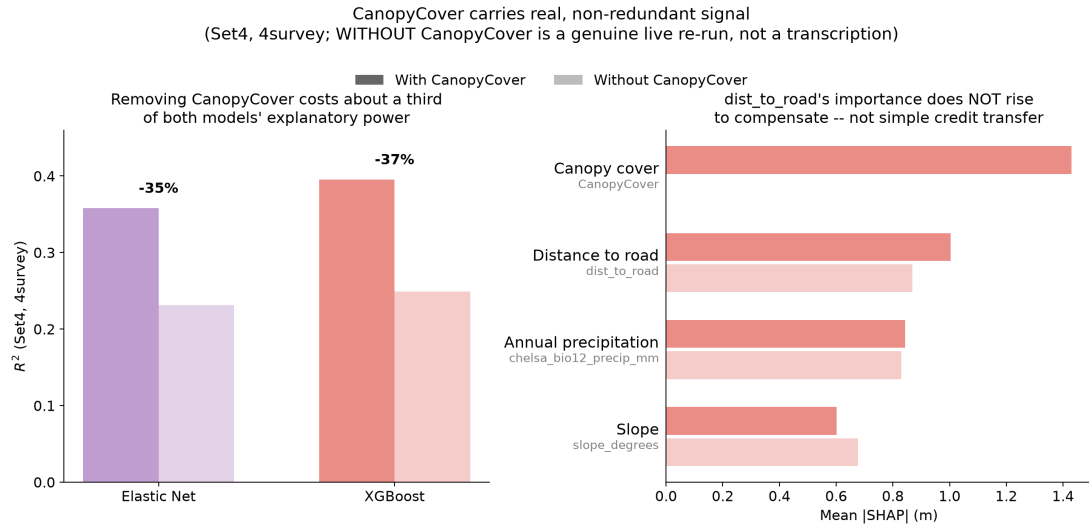


Figure 5.2: CanopyCover carries real, non-redundant signal. Left: pooled R^2 (Set4, 4-survey) with vs. without CanopyCover for Elastic Net and XGBoost, both lose roughly a third of their explanatory power. Right: mean |SHAP| for CanopyCover, *dist_to_road*, precipitation and slope, with vs. without CanopyCover, *dist_to_road*'s importance does not rise to compensate, evidence against simple correlation-driven credit transfer.

pools landscape-wide data. To find why coefficients disagree, we calculated within-compartment correlations by mean-centering predictors and targets per compartment, removing regional gradients to test if variables drive growth differences between neighbours.

This check sorts the six variables with sign disagreements (Figure 5.3, left. Only one, *topex* (green ring), retains a true local link: its correlation strengthens after removing compartment differences ($0.08 \rightarrow 0.21$), letting LMM recover a physical effect that pooled models obscure. Conversely, the other five (red rings) see correlations shrink towards zero wind speed (*gwa_weibull_a_50m*, $-0.33 \rightarrow -0.14$), temperature (*tas_mean*, $0.28 \rightarrow 0.09$), soil pH (*soilgrids_ph*, $0.22 \rightarrow 0.04$), wetness (*ceh_twi*, $-0.11 \rightarrow -0.01$), and local relief (*local_relief_500m*, $-0.12 \rightarrow -0.01$), showing raw correlations reflect compartment placement rather than local effects. Rainfall (*chelsa_bio12_precip_mm*, unringed) maintains agreement between models. Predictor instability thus stems from spatial confounding, not unmodelled nonlinearity.

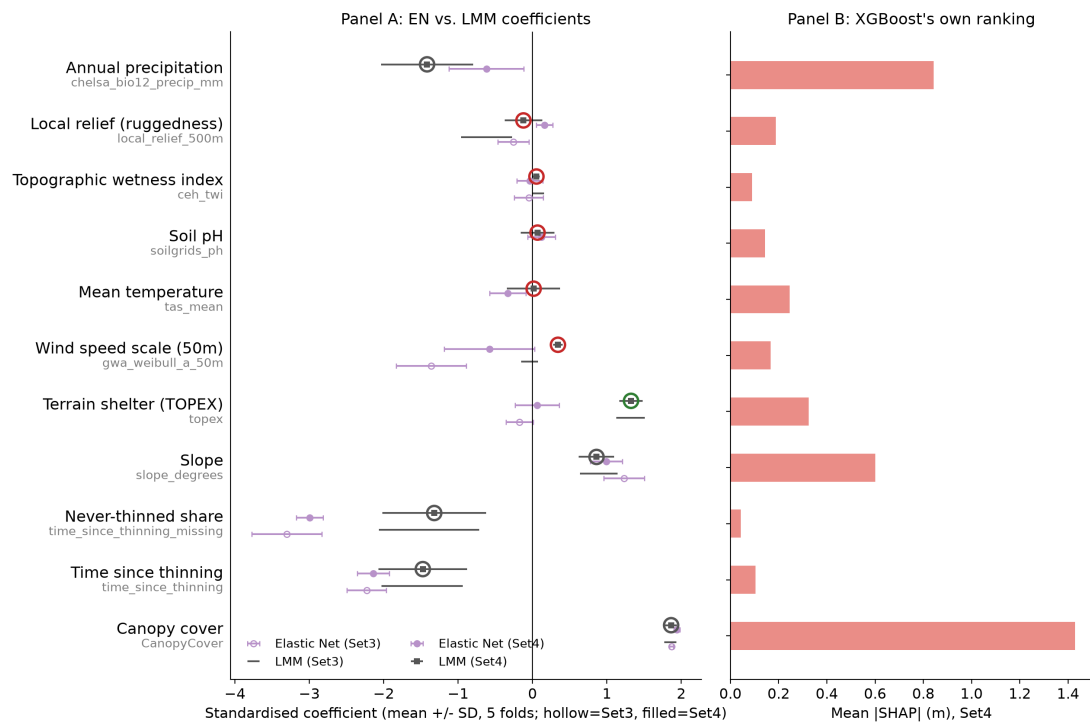


Figure 5.3: Comparing how different models rank and score environmental factors (Set 4, 4-survey cohort). Left: Model scores (average and range across 5 folds) for Elastic Net (purple) and Mixed-Effects (grey). Set 3 is shown with hollow markers, Set 4 with solid ones. For the six variables where the two models disagreed, green rings mark variables with a true local effect like terrain shelter, while red rings mark variables whose apparent effect disappears when looking within individual compartments. Rainfall is left unringed as a stable baseline. Right: XGBoost importance scores (SHAP) for the same variables.

Spatial clustering falls unevenly, and never fully disappears Every feature set still leaves real spatial clustering in its prediction errors (distance-band Moran's I at 1385–1739 m, $p = 0.001$; Figure 5.4B). Going from the stand-structure baseline (Set 1) to the full environmental model (Set 4), Elastic Net cuts this clustering by 29% ($0.145 \rightarrow 0.103$) and XGBoost by 63% ($0.146 \rightarrow 0.054$).

But XGBoost's drop is not smooth: clustering falls sharply at Set 2 ($0.146 \rightarrow 0.032$, a 78% drop), bounces back up at Set 3 (0.077), then falls again at Set 4 (0.054). This bounce comes from road distance (*dist_to_road*) and macroclimate variables (*chelsa_bio12_precip_mm*, *tas_mean*), which are missing from Set 3. Dropping these from Set 4 pushes clustering back up to 0.084 but costs almost no accuracy (R^2 only falls from 0.395 to 0.368). The clustering left in both models points to environmental drivers we have not measured, or spatial patterns that simply vary by place [53].

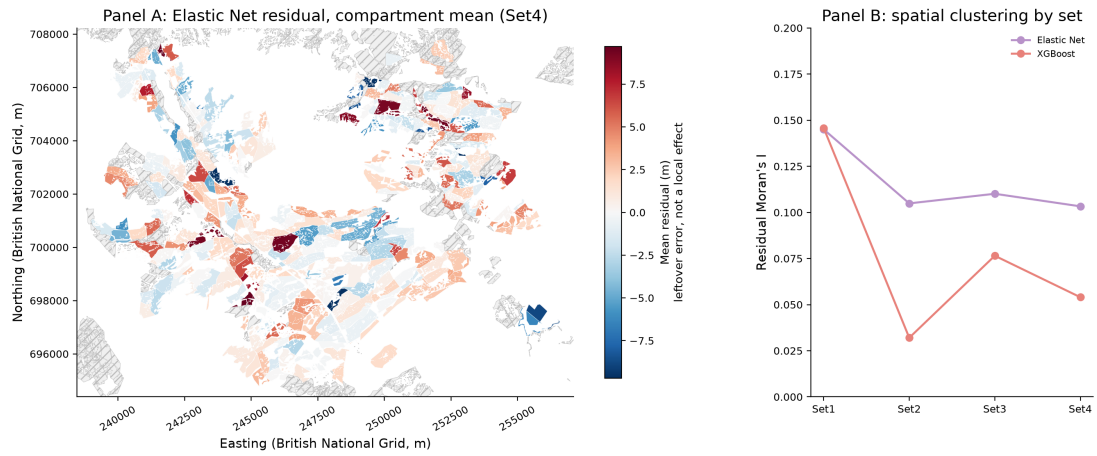


Figure 5.4: Spatial clustering of model residuals (Set 4, 4-survey cohort). Panel A (top): Set 4 Elastic Net residuals averaged by compartment, showing leftover spatial errors examined in Q2. Panel B (bottom): Residual Moran's I across feature sets. XGBoost reduces clustering more than Elastic Net overall, but drops sharply at Set 2 before bouncing back at Set 3.

Set 3 (terrain-only) is the weakest option. Refitting Set 4 category by category shows soil adds only a small gain ($+0.022 R^2$), while macroclimate actually hurts accuracy ($-0.056 R^2$; Appendix C). Terrain and edge-effect features drive most of Set 4's performance instead. This goes against the common assumption that forest yield modelling needs broad climate layers [54], here, regional climate grids add noise, not signal.

Every environmental model clearly beats the stand-structure baseline (Set 1), but XGBoost's confidence intervals for Set 2 and Set 4 overlap heavily ($[0.355, 0.461]$ vs. $[0.338, 0.438]$). So we cannot say with confidence that the full environmental set (Set 4) is really better than the simpler Set 2.

Q1: Conclusion

Stand structure and management history account for most plot-level growth variations. Among environmental factors, slope and rainfall remain stable across models, but six other variables show major disagreements between Elastic Net and the mixed-effects model. This happens because Elastic Net pools data across the whole landscape, turning regional artefacts into false signals; the mixed-effects model avoids this by looking within individual compartments, recovering true local effects for features like *topex* and wind exposure while exposing the others as regional noise. Overall, soil properties add almost nothing to predictions ($+0.022 R^2$), and macroclimate grids actively hurt accuracy ($-0.056 R^2$). Finally, significant spatial errors persist in all models' residuals,

meaning broad environmental data cannot fully explain regional patterns on its own. This leftover spatial error directly motivates the spatially varying coefficient models evaluated in Q2.

5.2 Q2: Spatial asymptote attribution

Q1 showed that global models leave strong spatial clustering in their prediction errors. Q2 tests whether a spatially varying model (GNNWR) can capture what these global models miss.

In terms of mean accuracy, GNNWR leads across every feature set (Table 5.2). On Set 4, it reaches $R^2 = 0.294$ ($[0.236, 0.347]$), ahead of Elastic Net (0.240 , $[0.192, 0.286]$) and XGBoost (0.250 , $[0.201, 0.295]$). However, broad overlap in confidence intervals means this lead is not clear-cut, and even its best score ($R^2 = 0.320$ on Set 2) leaves most growth variance unexplained. Looking into GNNWR's small advantage shows that because all three models rely almost entirely on canopy cover, GNNWR gains its edge simply by locally smoothing this single main feature, rather than finding subtle environmental trends missed by global baselines.

Method	Full Set4	No CanopyCover	CanopyCover only
Elastic Net	0.240 (0.223±0.100)	0.042 (0.024±0.077)	0.207 (0.190±0.084)
XGBoost	0.250 (0.231±0.119)	0.061 (0.040±0.094)	0.210 (0.193±0.081)
GNNWR	0.294 (0.277±0.068)	0.049 (0.026±0.067)	0.249 (0.235±0.030)

Table 5.2: Impact of canopy cover on model performance (Set 4, 4-survey cohort): Pooled R^2 (5-fold mean $\pm$ SD in parentheses). All three models collapse near zero without canopy cover ($R^2 < 0.07$). Canopy cover alone accounts for ~ 80 – 83% of each model's predictive power across both global baselines (Elastic Net, XGBoost) and the spatially-varying model (GNNWR).

5.2.1 Discussion

A noisy target explains the weak signal elsewhere Part of the reason other variables barely register is that Q2's target contains heavy noise. The fitting step fixes each plot's growth rate to its assigned yield class and estimates only the maximum height. When a plot grows faster or slower than assumed, that difference does not vanish; it spills straight into the estimated ceiling.

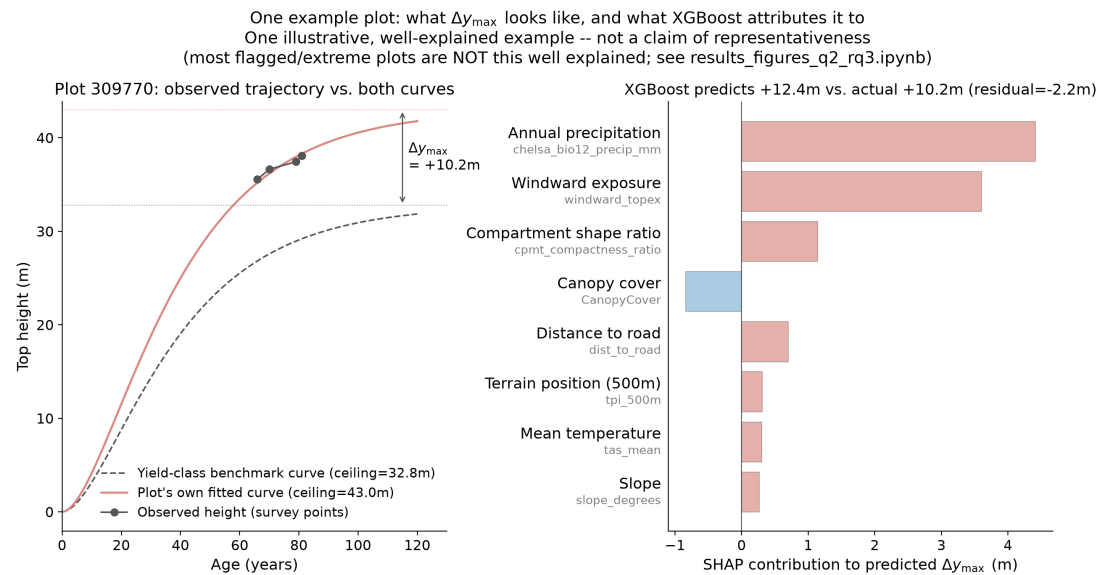


Figure 5.5: Example plot-level asymptote gap ($\Delta y_{\max}$) and XGBoost attribution. Left: Observed trajectory for Plot 309770, its fitted CR curve, and the yield-class reference curve, with the shaded region showing the observed gap across survey ages 65–81. Right: Local SHAP feature contributions for the same plot, illustrating how individual predictors account for the estimated ceiling departure.

Figure 5.5 shows this for Plot 309770, chosen because it is unusually well-explained, not typical: its fitted ceiling reaches 43.0 m, well above its reference value of 32.8 m (a +10.2 m gap), and XGBoost predicts +12.4 m, with canopy cover actually lowering the predicted ceiling here, unlike its usual positive effect. A typical plot, picked by median prediction error (Appendix Figure C.2), tells a different story: the model captures only about half the observed gap, with no single factor standing out, making this clean example the exception. Because $\Delta y_{\max}$ merges ceiling error and growth-rate mismatch into one number, only very strong signals like canopy cover stand out clearly, weaker, real environmental effects get lost, which is why most other site variables show little impact.

Q1 and Q2 agree on which variables to trust Q1 asks whether Elastic Net and the mixed-effects model (LMM) agree on the direction of a variable's effect (positive or negative). Q2 asks whether GNNWR's local coefficient points the same direction in every forest compartment, or flips between compartments. Canopy cover and slope pass both checks cleanly: Elastic Net and LMM agree on their direction in Q1, and in Q2 their GNNWR coefficients point the same way in almost every compartment (canopy cover 100%, slope 97% of the 231 compartments). Rainfall agrees on direction in Q1

too, and is mostly one-directional in Q2 (87% of compartments), though less uniformly so than canopy cover or slope. Soil pH fails both checks: Elastic Net and LMM disagree on its direction in Q1, and in Q2 its GNNWR coefficient is positive in barely half the compartments (51%) and negative in the rest, meaning no real, dependable relationship at either scale (Appendix Table C.2).

Topographic exposure (*topex*) is the interesting middle case. In Q1, Elastic Net and LMM disagree on its direction because *topex* has a real, fairly strong effect *within* each compartment (correlation 0.214) that looks much weaker once compartments are pooled together (correlation 0.080), so a model that only sees the pooled data gets the direction wrong. Q2 shows the same thing from a different angle: GNNWR's local coefficient for *topex* is positive in 72% of compartments but negative in the other 28% (Figure 5.6). Because Q1 and Q2 use entirely different statistical methods and still tell the same story, this instability looks like a real feature of the landscape, not an artefact of one particular model.

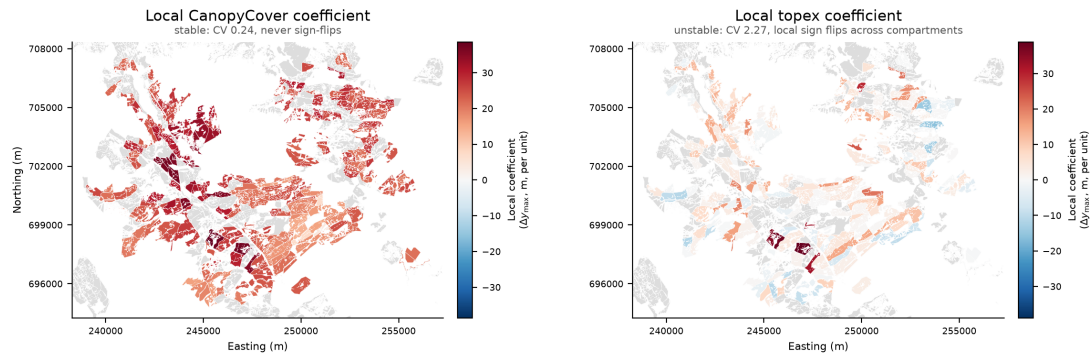


Figure 5.6: Spatial stability of GNNWR's local coefficients (Set 4, 4-survey cohort). Each map shows the compartment-average local coefficient on the same colour scale: how much $\Delta y_{\max}$ (m) changes per one-unit increase in the predictor. Left: canopy cover's coefficient stays positive in every compartment. Right: topographic exposure's (*topex*) coefficient is positive in most compartments but flips negative in over a quarter of them. This is why Elastic Net and LMM (Q1) disagree on *topex*'s direction: a global model forced to pick one sign for the whole landscape cannot capture this compartment-by-compartment flip.

A few minor limitations are worth noting. Some terrain features (like wind exposure and slope) show nonlinear patterns, rising before levelling off, which GNNWR's locally linear setup cannot fully capture. However, because canopy cover drives 80–90% of explained variance, this unmodelled nonlinearity has little practical impact. A tiny fraction of stands (under 1% of the 4-survey cohort; 266 of 56,526 plots) show extreme trade-offs between fitted growth rates and maximum heights (Appendix C). Finally, the

smaller 6-survey cohort produced near-zero accuracy across all models, reflecting its far smaller spatial sample (47 vs. 231 compartments).

Q2: Conclusion

Q2's result is a useful negative finding: letting the model vary its coefficients across space (GNNWR) did not reveal any hidden environmental driver beyond canopy cover, which the simpler global models had already found on their own. The diagnostics explain why. The target itself mixes two different kinds of error into one number, how fast a tree grows, and how tall it eventually gets, which makes any clean spatial pattern harder to find. GNNWR can only fit straight-line relationships within each local area, so it cannot capture terrain effects that curve or level off. In short, these are limits of the method and the target, not evidence that spatial ecology does not matter here.

5.3 Q3: Top-height prediction

Model	RMSE (m) ↓	MAE (m) ↓	R ² ↑
Linear	5.168 ± 0.361	4.056 ± 0.343	0.580 ± 0.026
RF	4.794 ± 0.187	3.611 ± 0.142	0.638 ± 0.011
XGBoost (tuned)	4.548 ± 0.137	3.431 ± 0.123	0.674 ± 0.009
DNN	4.681 ± 0.230	3.533 ± 0.201	0.655 ± 0.016
PINN*	4.842 ± 0.321	3.689 ± 0.289	0.631 ± 0.027
PINN- <i>k</i> *	4.927 ± 0.302	3.745 ± 0.271	0.618 ± 0.023

Table 5.3: Top-height model performance on Set 3 (4-survey cohort; mean ± SD across five spatial folds; $N = 232,292$ row-years across 58,073 plots). Bold indicates best performance. *PINN and PINN-*k* link predicted parameters directly to height predictions and are evaluated without hyperparameter tuning. Results for the 6-survey cohort are in Appendix C.4

Condition	PINN R^2	PINN- k R^2
<i>Feature set</i>		
No environment	0.572 ± 0.032	0.575 ± 0.033
Set 2 (small)	0.627 ± 0.023	0.622 ± 0.017
Set 3 (medium, baseline)	0.631 ± 0.027	0.618 ± 0.023
Set 4 (large)	0.618 ± 0.029	0.620 ± 0.020
<i>Hyperparameters (Set 3)</i>		
Main result	0.631	0.618
Learning-rate check	0.630	0.614
Batch-size check	0.641	0.619
λ check (8 settings)	0.623–0.632	0.612–0.620

Table 5.4: PINN and PINN- k test R^2 across feature sets and hyperparameter checks. Feature-set values report the 5-fold mean $\pm$ SD; hyperparameter values show single evaluation runs or the range across eight physics-weight (λ) values.

Metric	Plain PINN $\Delta y_{\max}$	PINN- k $\Delta y_{\max}$
Mean	+2.50 m	−0.45 m
SD	7.27 m	1.36 m
Max	+25.79 m	+4.95 m
Min	−30.45 m	−13.88 m
Implausible (< 5 m or > 70 m)	82 / 58,073 (0.14%)	0 / 58,073 (0%)

Table 5.5: Plausibility comparison of predicted maximum height deviations ($\Delta y_{\max}$) between plain PINN and PINN- k , pooled across all five folds ($N = 58,073$ plots). PINN- k produces zero implausible heights on the pooled dataset.

Discussion

XGBoost is the most accurate model overall ($R^2 = 0.674$), ahead of the standard DNN (0.655), PINN (0.631), and PINN- k (0.618; Table 5.3). It wins every fold on the 6-survey cohort and nearly every fold on the 4-survey data (trailing in just 1 of 5, by only 0.007 R^2). Because requiring six clean surveys leaves out steep, high-elevation stands (48 vs. 296 compartments), the rest of this chapter uses the broader 4-survey cohort. Four possible reasons for the DNN-to-XGBoost gap were tested and ruled out: feature interactions (DNN already captures 80% of the non-linear gain), training sample size (no wider gap when subsampling), network size (no gain from larger networks), and unusual terrain (XGBoost actually does worse than DNN there, error ratio 1.312 vs. 1.243). This sets up the key question for Q3: do PINNs’ safety and interpretability make up for their lower accuracy?

Environmental drivers are balanced, not dominated by one feature. We checked whether PINN- k leans on one hidden feature the way GNNWR leaned on canopy cover in Q2, using permutation importance (10 repeats per feature; Figure 5.7a). Here, influence is spread out: the top predictor (wind exposure) accounts for only 15.2% of total importance, with all eleven terrain and climate variables ranging from 4.7% to 15.2% (fold 0). On a typical stand (Plot 119937; Figure 5.7b), this spread-out adjustment still improves on the regional curve.

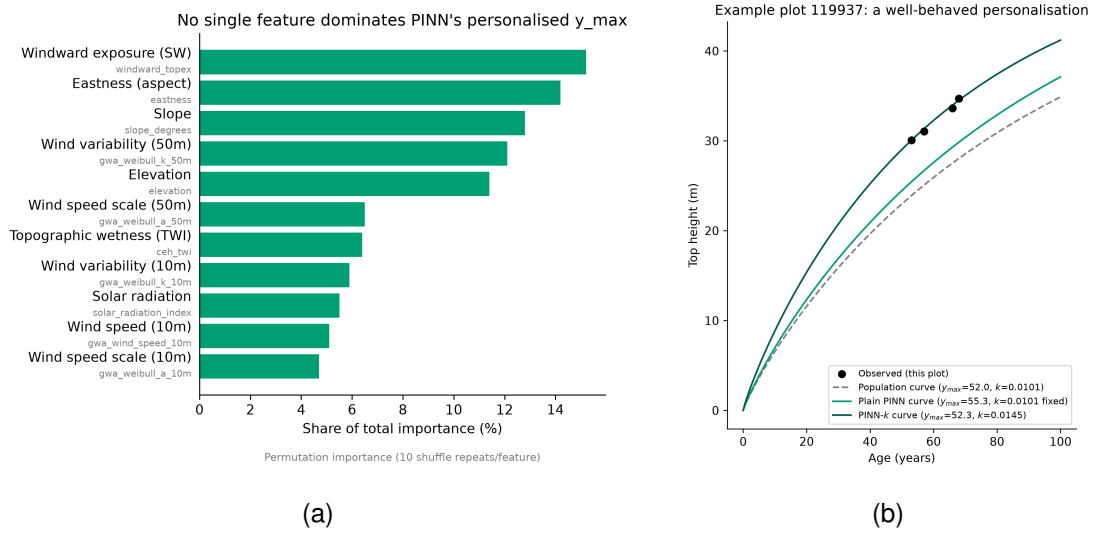


Figure 5.7: (a) Share of total permutation importance carried by each terrain/wind feature in PINN- k 's personalised y_{max} , fold 0. (b) Plot 119937 (compartment 2040, an ordinary compartment - rank 91 of 178, yield class 16, close to the population mean): both models' curves fit the observed points better than the population curve, PINN- k most closely of all. Not an extreme case, shown because personalisation should work sensibly on a typical plot, not just the flagged ones.

A small environmental feature set captures most of the signal We tested PINN on four environmental setups with 5-fold cross-validation: no site data, then three growing feature sets (Table 5.4, Figure 5.8). Even a small set (Set 2) gives an immediate jump over the no-environment baseline, in every fold, for both models. Bigger sets add nothing more, Set 2, Set 3, and Set 4 all score the same within fold-level noise. With site data, PINN- k is more consistent fold to fold than plain PINN; without it, both fall to the same baseline ($R^2 \approx 0.57$). So environmental data matters, but which set barely does; Set 3 is kept as the default mainly for consistency with the rest of the study.

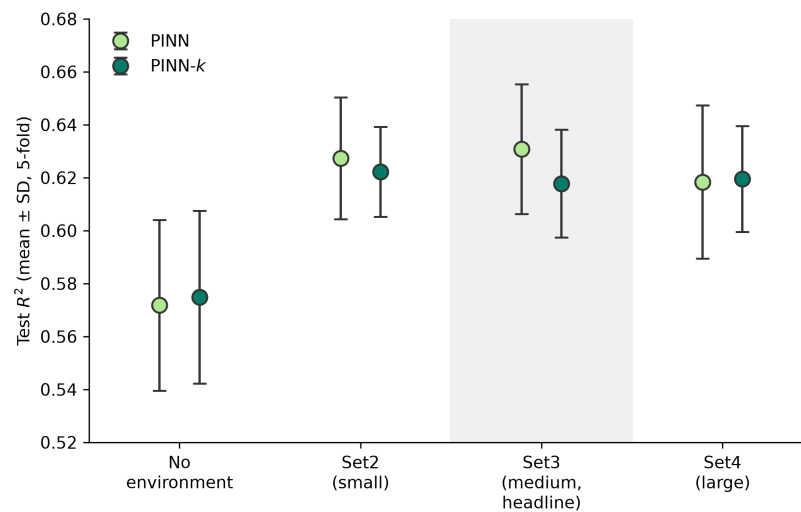


Figure 5.8: PINN and PINN- k accuracy against environmental feature-list size.

How PINN shifts curve parameters and matches yield classes Compartments were picked using a fixed ranking rule, not by eye (Appendix Table C.9). Among 178 compartments with ≥ 50 pooled test plots, Compartment 1129 had the single largest plot departure (+25.8 m; rank 1/178), and Compartment 2021 had both the largest mean shift and the largest negative departure (Figure 5.9). All compartments referenced here are shown on a master map (Appendix Figure C.4).

The top three over- and under-productive compartments agree closely between models (Appendix Figure C.5; Table C.8). But plain PINN and PINN- k route the adjustment differently (Figure 5.10): PINN- k keeps mean $y_{\max}$ shifts small (± 3 m) and pushes most of the signal through growth rate (k) instead, almost always upward. Four examples show this: Compartment 1045 adjusts mainly through $y_{\max}$, 2057 through k , 1070 through a negative $y_{\max}$ shift alone, and 2021 splits the adjustment across both.

To check biological plausibility, predictions were compared against yield-class records withheld from training. The five most under-productive compartments matched low yield classes in 4 of 5 cases; the five most over-productive matched high yield classes in 3 of 5. Across the full test set, predicted deviations correlate significantly with yield class (Appendix Table C.10; Appendix Figure C.6), showing the network picks up broad ecological patterns as well as local terrain effects.

Why some compartments disagree with yield class The over-productive stands that do not match their regional yield class follow a clear pattern (Appendix Table C.9): the two mismatched compartments (2057, 1027) are older (60 and 72 years) and

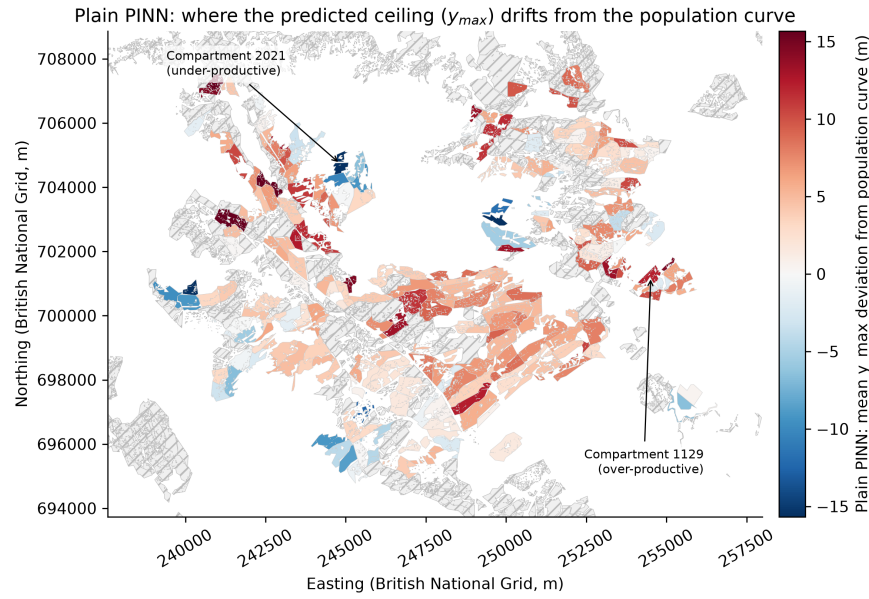


Figure 5.9: Compartment-level asymptote departures under standard PINN (pooled across 5 folds). Mean y_{max} deviation from the regional curve, highlighting flagship compartments 1129 and 2021.

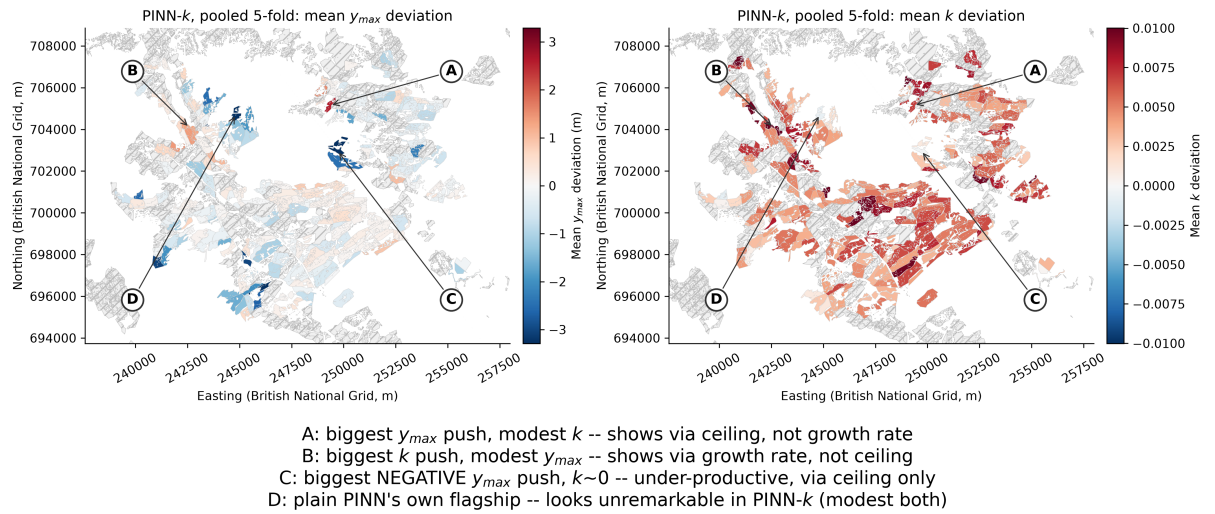


Figure 5.10: PINN- k , pooled across all 5 folds (58,073 plots): mean y_{max} deviation (left) vs. mean k deviation (right), by compartment. y_{max} stays close to flat almost everywhere (± 3 m); k shows a structured, almost entirely one-sided pattern instead, the same kind of signal, routed through a different parameter than plain PINN uses, and one that barely moves downward at all. Four compartments labelled A-D on both panels (legend below the figure) to show that different compartments route the signal differently, not just to repeat the single most extreme case four times.

wind-exposed, while the three matching ones are younger (25–35 years) and sheltered. Wind exposure is the strongest terrain driver overall (15.2% permutation importance; Figure 5.7a), but age alone is not the answer, PINNs are actually more accurate in older stands (Appendix Table C.7), so the real cause is still unclear. Compartment 2057 also has PINN- k 's single largest growth-rate deviation, meaning two independent model setups flag the same stand.

Biological plausibility differs sharply between PINN variants Plain PINN sometimes predicts extreme, biologically impossible heights for a small number of stands (Figure 5.11a). PINN- k never does this, staying realistic across all 58,073 test plots (Appendix Table 5.5, Figure 5.11b).

This happens because plain PINN can only adjust the height ceiling, so a strong environmental signal pushes that ceiling too high to fit fast early growth. PINN- k avoids this by putting that fast growth into the rate parameter (k) instead, leaving the ceiling stable. This is not just a fluke of cross-validation: even training directly on the affected compartments (four of five folds) still gave the same inflated ceiling (Table C.5). Whether these stands are genuinely exceptional or reflect old measurement errors is still an open question, but matching independent yield classes supports real site variation.

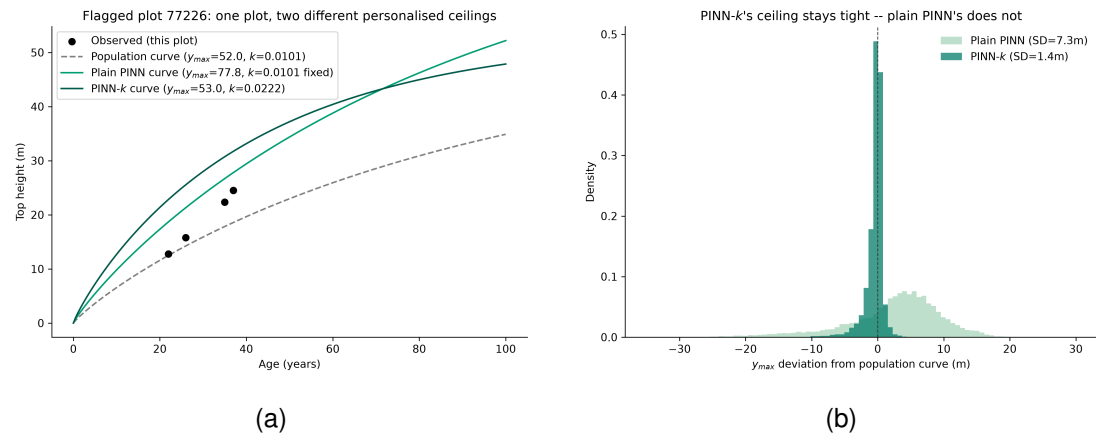


Figure 5.11: (a) Plot 77226 (compartment 1129): observed heights against the population curve, plain PINN's curve, and PINN- k 's curve. Plain PINN pushes the ceiling to an implausible 77.8m; PINN- k keeps the ceiling modest (53.0m) but personalises growth rate instead ($k=0.0222$, the dataset's single highest value), the two curves cross around age 70, PINN- k fitting the young observed points better before plain PINN's ceiling pulls ahead. (b) Distribution of y_{max} deviation from the population curve, plain PINN vs. PINN- k , pooled across all 5 folds (58,073 plots each). Same underlying signal, very different spread.

Model performance differs at age and height extremes Accuracy shifts across a stand's life (Table C.7). In young stands (0–10 m), the unconstrained DNN wins (MAE 2.68 m vs. 3.37 m for plain PINN, 3.15 m for PINN- k), since PINN's fixed curve shape struggles with early growth. In older stands (> 75 years, $n = 751$), it flips: the DNN is worst (6.83 m vs. 6.40 m for plain PINN), since nothing stops it drifting without a physical curve to anchor it. This matters for timber value: volume scales with height to the power 1.48–1.69, so a 10% height error becomes a 15–17% volume error. Since PINN's errors sit mostly in young stands, its volume predictions stay reliable where commercial value is highest.

Balancing predictive accuracy against biological safety Comparing plain PINN (height only) with PINN- k (height and growth rate) shows a clear trade-off: plain PINN is slightly more accurate in every setting tested (Table 5.4), but PINN- k models both growth parameters and never predicts an impossible height (Appendix table 5.5). Still, the two parameters move together sensibly ($r = 0.43$): positive growth-rate shifts pair with negative height shifts across compartments (Figure 5.10). Hyperparameter checks confirm plain PINN's edge holds across different settings (Table 5.4). Turning the physics loss off entirely ($\lambda = 0$) barely changes accuracy (PINN- k $R^2 = 0.620$ vs. 0.618 at the default $\lambda = 1$), with a small drop once $\lambda > 1$, physics guidance shapes *how* the model fits, not how well.

Q3: Conclusion

XGBoost is the most accurate model for top height, but physics-guided networks trade modest accuracy for safer predictions: PINN- k eliminates biologically impossible heights entirely, while plain PINN produces rare but extreme errors (82 of 58,073 plots, 0.14%; Table 5.5). Unlike Q2, Q3's environmental drivers are balanced across terrain and climate features, and performance varies by stand age: unconstrained DNNs lead in young stands, but PINN's fixed ceiling is more reliable in mature ones. As shown above, physics guidance does not boost raw accuracy, but it enforces biological plausibility (99.86% of plots for plain PINN, 100% for PINN- k) at a minor cost to precision.

Chapter 6

Conclusion and Future Work

6.1 Main conclusions

Canopy cover was the strongest correlate of departure from the shared growth curve (Q1). Management variables added smaller signals. Among the environmental variables, only *topex* and wind exposure showed repeatable local associations after compartment-level differences were considered. These are observational associations, not causal effects. Canopy cover should therefore not be interpreted as a proven management lever.

For the related $\Delta y_{\max}$ target, GNNWR provided only a small improvement over the global models (Q2). Canopy cover accounted for approximately 80–90% of its explained signal, based on the reduction in R^2 when it was removed. The added spatial complexity therefore revealed little further environmental structure.

XGBoost produced the most accurate top-height predictions (Q3). PINN- k had an RMSE approximately 0.4 m higher, but produced no predictions outside the predefined 5–70 m plausibility range. Plain PINN produced a small number of large violations. Physics guidance therefore improved plausibility rather than predictive accuracy. This finding applies only within the ages and forest conditions evaluated.

Together, the results address one part of the Forest Digital Twin concept: combining repeated observations and models to explain and predict forest change [1, 2]. Q1 and Q2 identify conditions associated with unusual growth. Q3 tests whether those signals support reliable height prediction. Real-time updating, disturbance simulation, management scenarios, and species beyond Sitka spruce remain outside the scope of this work.

6.2 Limitations

Data and targets The environmental layers were coarse relative to individual plots. SoilGrids has a 250 m resolution, while CHELSA and HadUK climate data are approximately 1 km [41, 44, 45]. These layers cannot represent fine-scale soil variation or short events such as storms. This may explain the weak and unstable soil and climate signals.

Historical uncertainty also affects the compartment records. Current species, yield-class, and thinning records were applied to earlier surveys and may not describe each compartment accurately at every date. Compartments 2057 and 1027, already flagged in Q3 for disagreeing with their assigned yield class, are exactly the kind of case this could produce, though this project cannot confirm whether their yield-class labels have actually changed over time. Q2's $\Delta y_{\max}$ target has a further limitation: with only four to six observations per plot, differences in growth rate and final height cannot be separated reliably. It is therefore a summary of trajectory departure, not a direct measurement of maximum height.

The six-survey cohort contains only 47 compartments, compared with 231 in the main cohort. Its weak performance may reflect limited spatial variation and unstable holdout folds rather than a fundamental model failure.

Models and evaluation Q1 used one five-fold spatial partition. Its findings, including the slope/*topex* stability contrast at the centre of Q1's Elastic Net/LMM disagreement, may therefore depend partly on the selected split. GNNWR was tested with additional partitions, but training randomness could not be separated from changes in fold membership, so its reported six-survey instability cannot be cleanly attributed to the model itself. Spatial blocking reduces leakage between nearby observations, but reliable evaluation still depends on having enough independent spatial groups [48, 13].

Elastic Net and the mixed-effects model assume linear relationships unless nonlinear terms are supplied. This assumption does not fully match the curved responses observed for slope and wind exposure, which qualifies Q1's finding that slope's coefficient sign agrees across models, agreement on sign does not rule out both models missing the same curvature, though this is likely a minor effect here since canopy cover dominates the explained variance. GNNWR was not systematically tuned, so its small advantage over the global models in Q2 remains uncertain. PINN and PINN- k were also evaluated on one cohort and one training run. The lambda ablation behind Q3's headline claim that physics guidance shapes fit rather than accuracy has not been checked on the six-survey

cohort or across seeds, and the separate contribution of the Chapman–Richards branch was not isolated.

6.3 Future work

The first priority is stronger evaluation. Future experiments should repeat all models across several seeds, tune GNNWR and PINN- k (each GNNWR run took roughly two hours, so cheaper equivalent models with a similar locally-varying structure are also worth testing), and test whether the Chapman–Richards branch contributes beyond the unconstrained neural component. Training on earlier surveys and testing on later surveys would measure temporal prediction, most naturally on the six-survey cohort, since it has the most years per plot, though that is also the cohort with too few compartments (48/296) for reliable spatial evaluation, so a temporal split would inherit the same small-sample problem, while evaluation in another forest, and in other commercially important species beyond Sitka spruce, would test geographic and species transfer.

Future attribution should also examine different forms of growth departure. Separate targets could represent persistently high or low growth, early acceleration, later plateauing, disturbance, and recovery. More surveys would help distinguish changes in growth rate from changes in height potential.

A fuller Forest Digital Twin could move from plot summaries to individual trees. Multi-temporal LiDAR can track tree-level growth and loss, although matching trees between surveys remains difficult [55]. Using complete three-dimensional point clouds could represent crown size, competition, structural damage, and mortality rather than top height alone [56].

Finally, finer soil and climate measurements could be combined with satellite embeddings such as AlphaEarth. These embeddings integrate spatial, temporal, and remote-sensing information, but would reduce direct interpretability and still require spatial and temporal validation [57].

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Appendix A

Appendix: Data

Field derivations and forestry definitions

Table A.1: Full derivation and forestry definitions for fields referenced in Table 3.2 but not spelled out there, one field per block. Units given wherever the source defines them.

Age	
Formula	$\text{LiDAR_year} - \text{plyr}$
Meaning	Stand age at the survey date, in years since planting
Used in project	Modelling input; cleaning filter. Field-description doc states valid range 0–150 years; <code>clean_master_data.py</code> implements 0–200, documentation/code mismatch, unresolved
Top_Height95	
Formula	$\text{elev_percentile_95th} \times 1.1$
Meaning	FR's corrected estimate of Top Height (m): mean height of the 100 largest-diameter trees per hectare. The $\times 1.1$ factor is FR's stated correction for known underestimation at the 95th return percentile
Used in project	Audit ingredient only (feeds <code>Vol95/GYCspec95</code>). This project's target is the <i>uncorrected</i> <code>elev_percentile_95th</code> , a deliberate deviation from FR's own corrected convention
Vol95	
Formula	$(g_1 \cdot \text{Top_Height95}^{g_3}) \times (\text{area}/10000) \times \text{CanopyCover}$
Meaning	Stand volume (m^3), scaled from a per-ha allometric relationship by plot area (ha) and canopy cover fraction
Used in project	Used downstream as a disturbance-screening signal. g_1/g_3 are pre-assigned per plot in the source FR export, not computed or selected by this project, e.g. $g_1=3.509144$, $g_3=1.478136$ and $g_1=2.922183$, $g_3=1.694732$ are two real examples.
Vol_RM95	
Formula	$(-35.8733 + 5.3486 \cdot \text{Top_Height95}^{1.5424}) \times (\text{area}/10000) \times \text{CanopyCover}$
Meaning	Robertson & Miller generic volume equation (m^3), species-independent alternative to <code>Vol95</code>
Used in project	Not kept in the master export; not used
GYCspec95	
Formula	$(\text{Top_Height95} / (1 - e^{-p_4 \cdot \text{Age}}))^{p_5} - p_1 - p_3 \times 2 / p_2$
Meaning	General Yield Class (continuous scale; FR's own tables report GYC in multiples of 2, e.g. YC14, YC16); potential productivity, the maximum mean annual increment of cumulative volume ($\text{m}^3/\text{ha}/\text{yr}$) for the species on that site under standard management

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Used in project	Never used downstream; would be leakage if it were (contains Top_Height95)
y_{max}^{ylc} (RQ3)	
Formula	$ylc \cdot p_2 + p_1 + p_3 \times 2$
Meaning	The height ceiling (m) that the plot's independently assigned yield class implies, under the same site-index model as GYCs _{spec95} , solved the opposite direction
Used in project	Used, RQ3's independent comparison baseline (data_processing/export_growth_curve_tables.py)
DBH_{mean} source PDF/report, not yet in citation list	
Formula	$(b_1 + b_2 \cdot \text{Spacing}) \times \text{TopHeight}^{b_4}$
Meaning	Mean diameter at breast height, 1.3 m above ground (cm); needed to estimate stems per hectare
Used in project	Not present anywhere in this project's data or code. Supporting context only, for the claim that top height feeds an established forestry allometric toolkit (SS, thinned: $b_1=0.714157$, $b_2=0.239399$, $b_4=1.079545$; unthinned: $b_1=1.212457$, $b_2=0.609335$, $b_4=0.754978$)
BA same source	
Formula	$h_1 \times \text{TopHeight}^{h_4}$
Meaning	Basal area (m ² /ha): cross-sectional stem area summed per hectare
Used in project	Not present in this project's data. Same supporting-context caveat (SS, thinned: $h_1=7.499437$, $h_4=0.497925$; unthinned: $h_1=5.517392$, $h_4=0.771801$)
Volume (source PDF, Eq. 13) same source	
Formula	$g_1 \times \text{TopHeight}^{g_3}$ (no area/canopy scaling)
Meaning	Volume over bark, per hectare (m ³ /ha)
Used in project	Distinct equation from this project's own Vol ₁₉₅ above, same g_1/g_3 notation, different formula (no area/CanopyCover scaling). Do not conflate the two in the appendix text
Top Height (forestry definition)	
Formula	,
Meaning	Mean height of the 100 largest-diameter trees per hectare (m)
Used in project	elev_percentile_95th is an ALS-derived proxy for this classical mensuration concept, not identical to it by construction
Yield Class (forestry definition)	
Formula	,
Meaning	Index of site productivity: max mean annual increment of cumulative volume (m ³ /ha/yr); tabulated in multiples of 2 in FR yield tables, continuous here
Used in project	ylc is this project's realisation of it

Environmental variable definitions and coverage

Table A.2: Per-variable definitions, extraction method and coverage for the environmental table (71,766 plots). Source: methodology_env_setpick.md, verified 2026-08-10 against plot_environmental_features.parquet. Feature-selection, VIF screening and which variables entered a final model set are described in Chapter 4, not here.

Variable	Family	Extraction method	Valid/ Missing	Distinct	Min–Max
elevation	Terrain	Direct raster read	71,766/0	4,028	18.2–561.1 m
slope_degrees	Terrain	$\arctan(\sqrt{dz_{dx}^2 + dz_{dy}^2})$, np.gradient	71,766/0	15,326	0.0–42.84°
northness, eastness	Terrain	cos / sin(aspect, radians)	71,766/0	15,387/15,572	–1–1
profile_/plan_	Terrain	2nd derivative of DTM zevenber-	71,766/0	16,835/16,829	–0.007–0.008 / –0.67–
curvature		gen1987			0.40

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Variable	Family	Extraction method	Valid/ Missing	Distinct	Min–Max
local_relief_500m	Terrain	Max–min elevation, 21×21-cell window	71,766/0	6,086	7.5–461.9 m
solar_radiation_index	Terrain	cos(slope) cos(zenith) + sin(slope) sin(zenith) cos(aspect – azimuth), no horizon shading	71,766/0	16,012	0.338–1.000
frost_hollow_flag	Terrain	Binary: TPI < own 15th ptile AND plan curvature < 0	71,766/0	2	0 ($n=62,691$)/1 ($n=9,075$)
ceh_twi	Terrain	Direct raster read (CEH TWI)	71,727/39	16,800	3.75–16.86
topex	Wind	Sum of horizon angles, 8 directions wilson1984	71,766/0	16,839	–85.9–68.6°
windward_topex	Wind	Same, single 225° bearing	71,766/0	8,093	–19.8–19.4°
gwa_weibull_a/k_10m/50m	Wind	Direct raster read, national grid	71,766/0	1,816–1,824	varies by variable
gwa_wind_speed_10m	Wind	Nearest-cell extraction, REST API	71,766/0	1,816	0.069–11.18 m/s
whcl	Wind	Direct field read (raw GPKG)	71,766/0	6 of 7 possible	Class 1 absent
soilgrids_ph	Soil	Direct read, topsoil 0–5 cm	71,353/413	13	4.0–5.2
ceh_pedotope	Soil	Categorical, one-hot	71,727/39	7 levels	,
ceh_subsurface_drainage	Soil	Categorical, one-hot	71,727/39	3 levels	,
ceh_textural_composition	Soil	Categorical, one-hot	71,727/39	4 levels	,
dist_to_watercourse	Soil/spatial	Distance to nearest OS Open Rivers line	71,766/0	71,758	0.0045–1850.9 m
chelsa_gdd5_degc	Climate	Direct read, 1981–2010	71,766/0	206	906.5–1580.8
chelsa_bio12_precip_mm	Climate	Direct read, 1981–2010	71,766/0	269	1729.8–3004.8
tas_mean, groundfrost_mean	Climate	Cohort-suffixed, resolved to cohort's real years	71,766/0	149 each	5.75–9.15 / 8.76–9.47
dist_to_scpt/_/_block_boundary	Spatial	Distance to own sub-cpmt/block polygon	71,766/0	68,188/53,974	~0–217 m
dist_to_forest_perimeter	Spatial	Distance to buffered union of compartments	71,766/0	71,701	~0–764.7 m
cpmt_compactness_ratio	Spatial	Perimeter/area, per compartment	71,766/0	296	0.0097–0.240
dist_to_road	Spatial	Distance to nearest OS Open Roads line	71,766/0	71,580	0.056–2375.7 m
CanopyCover	Stand	Mean fraction across survey years	71,766/0	70,769	0.008–0.996
time_since_thinning	Stand	NaN-filled with 0 downstream	71,766/0	19	0–16 yr

External data source details

Table 3.3 in the Data chapter summarises the seven data families used; this table gives the access method, licensing, and temporal vintage behind each one.

Data family	Provider and access	Licence	Temporal vintage and caveats
Terrain	OS Terrain 50 DTM, direct raster read	OS OpenData	Single static snapshot; not matched to individual survey years
Soil	CEH Natural Capital pedotope raster (one of 9 layers in a single downloaded zip); SoilGrids v2.0 (ISRIC, public Cloud-Optimised GeoTIFF via rasterio's /vsicurl/, no authentication)	CEH: institutional download; SoilGrids: public, no auth	Static; SoilGrids pH uses only the shallowest of six depth bands (0–5 cm topsoil)
Wind exposure	Global Wind Atlas GBR national rasters (Weibull-A/K, mean wind speed), direct raster read and public REST API	Public	2006–2015 wind climatology: one fixed value per plot location, not recalculated per survey year
Observed wind	MIDAS Open weather stations	Public (Met Office)	Computed per survey-to-survey interval from hourly station observations, not a static layer
Long-term climate	CHELSEA v2.1, direct public download, no authentication	Public	1981–2010 climatology (terrain-conditioned downscaling, not plain interpolation); static, not tied to any survey year
Matched climate	HadUK-Grid, CEDA	Public	Genuinely multi-year: averaged across each cohort's own real survey years (2008/2012/2021/2023 for the 4-survey cohort; all six years for the 6-survey cohort), not a single reused snapshot
Spatial context	OS Open Roads and Open Rivers, vector line data	OS OpenData	No stated vintage in this project's own downloaded package docs; the raw file's download timestamp is ~July 2026, applied retroactively to every survey year back to 2002. Minor for rivers (stable over time); more relevant for roads, since forestry access tracks can be built within that timeframe

Table A.3: External data source access method, licensing, and temporal vintage, one row per data family in Table 3.3.

Appendix B

Appendix: Methodology

Feature-set membership by research question Ranking and screening is repeated separately for each question’s own target, so final Set membership differs even where two questions share a Set number. Sets are not strictly nested: VIF-based collinearity screening (variables with variance inflation factor > 5 removed iteratively) is re-run each time the candidate pool expands, so a variable present in a smaller Set can be screened out once a larger Set’s wider pool makes it redundant. Elevation is the clearest example: it survives Set2 and Set3 in every question below, but is removed from Set4 once the wider terrain/wind pool is considered together, this reflects statistical redundancy with other terrain/wind variables (elevation is a well-known confound for temperature, wind exposure and radiation), not that elevation is ecologically unimportant.

Q1 (target mean_cr_residual)

Set	<i>n</i>	Columns
Set1	4	CanopyCover, time_since_thinning, time_since_thinning_missing, recent_thinning_5yr
Set2	14	Set1 + chelsa_bio12_precip_mm, chelsa_gdd5_degc, gwa_weibull_k_50m, slope_degrees, gwa_weibull_a_10m, local_relief_500m, eastness, dist_to_road, gwa_weibull_a_50m, tas_mean
Set3	14	Set1 + gwa_weibull_k_50m, slope_degrees, gwa_weibull_a_10m, local_relief_500m, eastness, gwa_weibull_a_50m, ceh_twi, whcl, topex, gwa_wind_speed_10m
Set4	19	Set3 + chelsa_bio12_precip_mm, dist_to_road, tas_mean, dist_to_scpt_boundary, soilgrids_ph
Set5	28	Set1 + slope_degrees, northness, eastness, profile_curvature, plan_curvature, tpi_500m, local_relief_500m, frost_hollow_flag, windward_topex, gwa_weibull_a_10m, gwa_weibull_k_10m, gwa_weibull_a_50m, gwa_weibull_k_50m, dist_to_forest_perimeter, dist_to_block_boundary, cpmt_compactness_ratio, dist_to_road, dist_to_watercourse, gwa_wind_speed_10m, soilgrids_ph, ceh_twi, tas_mean, groundfrost_mean, whcl

Table B.1: Q1 (RSQ2) feature-set membership, elastic_net/XGBoost/LMM attribution pipeline.

Q2 (target local_y_max_difference)

Set	<i>n</i>	Columns
Set1	4	CanopyCover, time_since_thinning, time_since_thinning_missing, recent_thinning_5yr
Set2	14	Set1 + windward_topex, elevation, cpmt_compactness_ratio, gwa_wind_speed_50m, slope_degrees, tpi_500m, dist_to_road, topex, solar_radiation_index, soilgrids_ph
Set3	12	Set1 + windward_topex, elevation, gwa_wind_speed_50m, slope_degrees, tpi_500m, topex, solar_radiation_index, northness
Set4	20	Set1 + windward_topex, gwa_wind_speed_50m, slope_degrees, tpi_500m, topex, northness, cpmt_compactness_ratio, dist_to_road, chelsa_bio12_precip_mm, tas_mean, soilgrids_ph, ceh_subsurface_drainage=2.0, ceh_pedotope=2.0, ceh_pedotope=8.0, ceh_textural_composition=2.0, dist_to_forest_perimeter
Set5	31	Set1 + slope_degrees, northness, eastness, profile_curvature, plan_curvature, ceh_twi, frost_hollow_flag, windward_topex, whcl, tpi_500m, local_relief_500m, tas_mean, groundfrost_mean, soilgrids_ph, dist_to_watercourse, dist_to_forest_perimeter, dist_to_scpt_boundary, cpmt_compactness_ratio, dist_to_road, ceh_pedotope=11.0, ceh_pedotope=2.0, ceh_pedotope=5.0, ceh_pedotope=8.0, ceh_pedotope=9.0, ceh_subsurface_drainage=2.0, ceh_subsurface_drainage=3.0, ceh_textural_composition=5.0

Table B.2: Q2 (RSQ3) feature-set membership, GNNWR spatial-attribution pipeline. Note Set3's non-monotonic size (12 columns, smaller than Set2's 14) is expected, not an error: Set2 and Set3 are built by two different rules (top-10 by rank vs. top-half-of-category by rank) over overlapping but not identical candidate pools, Set3 is not defined as a superset of Set2.

Q3 (target elev_percentile_95th)

Set	<i>n</i>	Columns
Set1	4	CanopyCover, time_since_thinning, time_since_thinning_missing, recent_thinning_5yr
Set2	14	Set1 + gwa_weibull_k_50m, dist_to_scpt_boundary, elevation, tas_mean, eastness, windward_topex, gwa_weibull_a_10m, gwa_weibull_a_50m, slope_degrees, cpmt_compactness_ratio
Set3	15	Set1 + gwa_weibull_k_50m, elevation, eastness, windward_topex, gwa_weibull_a_10m, gwa_weibull_a_50m, slope_degrees, gwa_weibull_k_10m, solar_radiation_index, ceh_twi, gwa_wind_speed_10m
Set4	18	Set1 + gwa_weibull_k_50m, eastness, windward_topex, gwa_weibull_a_10m, gwa_weibull_a_50m, slope_degrees, gwa_weibull_k_10m, solar_radiation_index, ceh_twi, gwa_wind_speed_10m, dist_to_scpt_boundary, tas_mean, chelsa_gdd5_degc, cpmt_compactness_ratio
Set5	30	Set1 + slope_degrees, northness, eastness, profile_curvature, plan_curvature, local_relief_500m, frost_hollow_flag, topex, windward_topex, gwa_weibull_a_10m, gwa_weibull_k_10m, gwa_weibull_a_50m, gwa_weibull_k_50m, dist_to_forest_perimeter, dist_to_scpt_boundary, cpmt_compactness_ratio, dist_to_road, dist_to_watercourse, gwa_wind_speed_10m, soilgrids_ph, ceh_twi, chelsa_gdd5_degc, chelsa_bio12_precip_mm, tas_mean, groundfrost_mean, whcl

Table B.3: Q3 (RSQ1) feature-set membership, DNN/PINN/PINN-*k* prediction pipeline.

Feature-set purpose by question

Feature set	Included feature groups	Model comparison	Environmental attribution	Reason
Q3 (RSQ1) Set1–Set5	Baseline + terrain/wind/climate/soil/edge (VIF-screened)	Yes (primary)	No	Feeds <code>dnn_env_terrain/pinn_env_terrain*</code> , designed for a fair DNN-vs-PINN comparison under progressively richer environmental input, and to test whether environmental conditioning improves height prediction. No linear/coefficient model reads these; not designed for attribution.
Q1 (RSQ2) Set1–Set5	Baseline + terrain/wind/climate/soil/edge (VIF-screened)	Secondary (Elastic Net vs. XGBoost vs. LMM/NLME)	Yes (primary)	Feeds Elastic Net/XGBoost/LMM on <code>mean_cr_residual</code> , designed to identify which conditions align with persistent departure from the shared CR curve. Model comparison across the three fitted models is possible but secondary to the attribution question.
Q2 (RSQ3) Set1–Set5	Baseline + terrain-wind/climate/soil-site/edge.position (VIF-screened, one-hot, reference level dropped)	Secondary (Elastic Net vs. XGBoost vs. GNNWR)	Yes (primary)	Feeds Elastic Net/XGBoost/GNNWR on <code>local_y_max_difference</code> , per-plot attribution; GNNWR specifically tests whether attribution is spatially varying, not just globally linear.
Q2 (RSQ3) SCOPE_GROUPS (pre-existing, untouched)	Named category combinations (terrain_wind, +climate, +soil_site, +edge_position, +management, broad_environment, broad_environment_plus_management)	No	Yes (category-level)	A <i>different</i> , coarser question, “does adding this whole category help”, answered by a pre-existing system this work did not modify. No Q1 equivalent of this specific system exists, though Q1 answers the same coarser question via its own separate category-ablation method (see Q1 appendix).
6-survey informational check (Q1/RSQ2 Set2 only)	Same candidate pool, 6-survey cohort	No	Diagnostic only	Not a primary result, checks whether Set2’s top-10 changes on the smaller (7,467-row), noisier 6-survey cohort. It does (only 1 of 10 candidates overlap with 4-survey’s Set2), reported as a caveat on Set2’s stability, not as a second deliverable.

Table B.4: What each feature set is for, and which question(s) it serves. Widening a Set (Set2→Set3→Set4→Set5) is not claimed to necessarily improve prediction or strengthen an attribution finding, each Set is a candidate for downstream evaluation, and which one performs best is an empirical question for the project’s real five-fold spatial CV, not decided by this screening pipeline.

Appendix C

Appendix: Results

Q1 appendix

Full Set4 variable table (EN vs. LMM, static/dynamic, stability)

Table C.1: Variance inflation factor (VIF) for all 19 Set4 columns (4-survey), computed live via `compute_vif_table()`. Only the 3 thinning/management columns exceed the standard VIF=5 cutoff (expected, `time_since_thinning` and its missingness flag are closely related by construction); all abiotic (terrain/climate/soil) variables discussed in the main text, including the 6 unstable ones, have VIF well under 5, ruling out multicollinearity as their cause. `CanopyCover` itself also has low VIF despite being the strongest predictor of the target: VIF measures how well a variable is explained by the *other* predictors, not how strongly it predicts the target, and `CanopyCover` (a stand-structure attribute) is not well explained by the terrain, wind, climate, and soil columns around it. This is the same conclusion as the earlier non-redundancy check (Figure 5.2) restated in different terms: `CanopyCover` carries information the other 18 columns cannot substitute for.

Variable	EN (mean±SD)	coef. LMM (mean±SD)	coef. Same sign?	Static/Dynamic	Stability dict	ver- Basis
CanopyCover	+1.943 0.038	± +1.864 0.078	± Yes	Dynamic	Not assessed , (leakage check instead)	
time_since_thinning	-2.139 0.216	± -1.475 0.596	± Yes	Dynamic	Not assessed ,	
time_since_thinning_- missing	-2.995 0.182	± -1.321 0.698	± Yes	Dynamic	Not assessed ,	
recent_thinning_5yr	-0.975 0.273	± -0.403 0.286	± Yes	Dynamic	Not assessed ,	
slope_degrees	+0.993 0.217	± +0.861 0.238	± Yes	Static	Stable	Text-discussed

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Variable	EN (mean±SD)	coef. LMM (mean±SD)	coef. Same sign?	Static/Dynamic	Stability ver- dict	Basis
topex	+0.063 0.296	± +1.325 0.155	± Yes only; flips Set3→Set4)	Static	Unstable	Text-discussed; real local signal (0.09→0.21)
tas_mean	-0.334 0.244	± +0.016 0.358	± No	Dynamic	Unstable	Text-discussed; artifact (0.27→0.08)
soilgrids_ph	+0.123 0.186	± +0.068 0.225	± Yes only)	Static	Unstable	Text-discussed; artifact (0.27→0.05)
chelsa_bio12_precip.- mm	-0.620 0.505	± -1.417 0.618	± Yes	Static	Stable (EN/LMM agree)	Text-discussed; local-signal retention (-0.298 → -0.0996, 33%) is close to the confirmed-artifact tas.-mean's (32%), stability here rests on EN/LMM agreement itself, not a clearly demonstrated surviving local signal like topex's (267%)
ceh_twi	-0.038 0.174	± +0.053 0.047	± No	Static	Unstable (new)	Found this session; artifact (-0.10 → 0.01)
gwa_weibull_a_50m	-0.579 0.609	± +0.342 0.066	± No	Static	Unstable (new)	Found this session; real local signal (-0.34 → -0.14)
local_relief_500m	+0.163 0.110	± -0.122 0.254	± No	Static	Unstable (new)	Found this session; artifact (-0.15 → 0.00)
gwa_weibull_k_50m	-0.135 0.196	± -0.359 0.249	± Yes	Static	Not assessed	,
gwa_weibull_a_10m	-0.082 0.094	± -0.111 0.056	± Yes	Static	Not assessed	,
gwa_wind_speed_10m	-0.121 0.074	± -0.108 0.068	± Yes	Static	Not assessed	,
eastness	+0.471 0.087	± +0.326 0.019	± Yes	Static	Not assessed	,
dist_to_scpt_boundary	-0.299 0.103	± -0.146 0.071	± Yes	Static	Not assessed	,
dist_to_road	-1.332 0.171	± -1.216 0.130	± Yes	Static (vintage uncertain)	Not assessed (own role in leakage check)	,
whcl	-0.594 0.228	± -0.459 0.400	± Yes	Static (not verified)	Not assessed	,

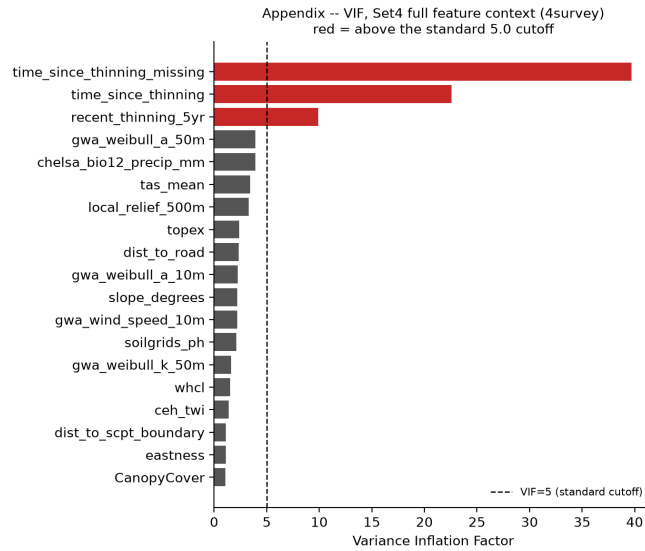


Figure C.1: Variance inflation factor (VIF) for all 19 Set4 columns (4-survey), computed live via `compute_vif_table()`. Only the 3 thinning/management columns exceed the standard VIF=5 cutoff (expected, `time_since_thinning` and its missingness flag are closely related by construction); all abiotic (terrain/climate/soil) variables discussed in the main text, including the 6 unstable ones, have VIF well under 5, ruling out multicollinearity as their cause.

Category-level ablation (soil / climate / terrain) Removing soil variables (`soilgrids_ph`, `ceh_pedotope`, `ceh_subsurface_drainage`, `ceh_textural_composition`, `dist_to_watercourse`) from Set4 changes test R^2 by $+0.022$; removing macroclimate variables (`tas_mean`, `groundfrost_mean`, `chelsa_bio1_celsius`, `chelsa_gdd5_degc`, `chelsa_bio12_precip_mm`) changes it by -0.056 , i.e. climate variables reduce held-out accuracy when present. These two headline deltas are the figures already quoted in the main Q1 Discussion. The full refit table behind them (Set4 vs. each category individually removed, all five categories) has not been re-verified against a saved source file this session; a fuller breakdown table will be added here once that source run is reproduced or reconciled with the mismatched candidate noted above.

Terrain/climate variable reliability, Q1 vs. Q2

Variable	Q1: EN vs. LMM	Q1 mechanism (if disputed)	Q2: % Read cpmt. same sign	
CanopyCover	Agree (EN 1.94, LMM 1.86)	not disputed	100.0	Reliable everywhere
slope_degrees	Agree (EN 0.99, LMM 0.86)	not disputed	97.0	Reliable everywhere
chelsa_bio12_precip_mm	Agree (EN -0.62, LMM -1.42)	not disputed	87.4	Reliable everywhere
topex	Disagree (EN 0.06, LMM 1.32)	Real local signal (within-cpmt. corr. 0.214 > raw 0.080)	71.9	Real effect, hard for a straight line (global or local) to pin down consistently
windward_topex	not tested in Q1	n/a	91.8	Q2-only variable, more stable than plain topex
tas_mean	Disagree (EN -0.33, LMM 0.02)	Artifact (within-cpmt. corr. 0.090 << raw 0.279)	78.4	Q1: fake signal, despite a fairly stable local sign in Q2, stability alone does not make it a real effect
soilgrids_ph	Disagree (EN 0.12, LMM 0.07)	Artifact (within-cpmt. corr. 0.035 << raw 0.224)	51.1	Both chapters agree: not a real effect
dist_to_road	not tested in Q1	n/a	74.9	No Q1 comparison; moderately stable sign in Q2

Table C.2: Terrain/climate variable reliability, cross-referenced between Q1 (does a global Elastic Net model agree with a linear mixed-effects model on a variable's coefficient, Set4, 4-survey) and Q2 (does GNNWR's own local coefficient keep the same sign across compartments, Set4, 4-survey, pooled 5 folds, $n = 231$ compartments). These are two genuinely different tests that happen to agree on topex and slope_degrees, not the same measurement repeated.

A typical example plot

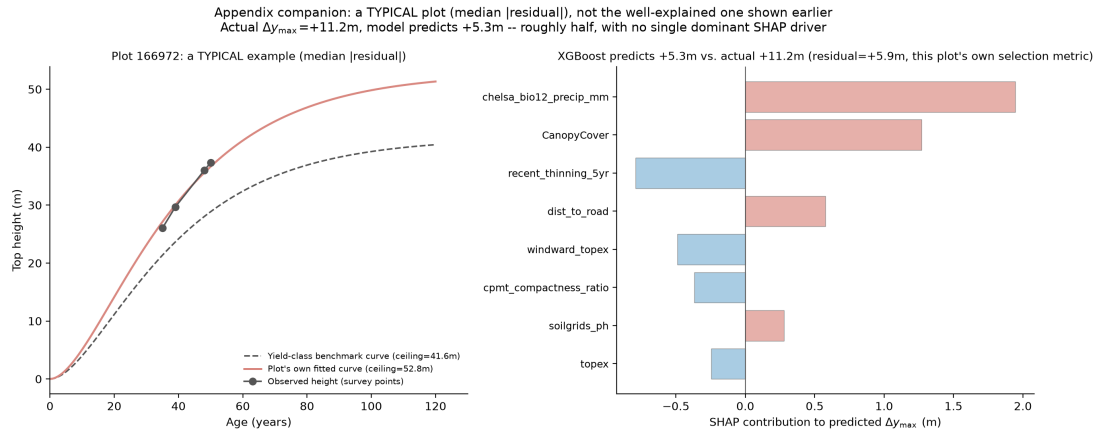


Figure C.2: Plot 166972, chosen because its prediction error sits at the population median, not the well-fit tail (ages 35–50, heights 26.0–37.3 m). Left: observed trajectory against its own fitted curve and the yield-class benchmark curve. Right: XGBoost SHAP contributions to the predicted $\Delta y_{\max}$. The actual deviation is $+11.2\text{m}$ but XGBoost predicts only $+5.3\text{m}$, roughly half, with a diffuse SHAP breakdown (*chelsea_bio12_precip_mm*, *CanopyCover*, *recent_thinning_5yr*, *dist_to_road*, *windward_topex*, none dominant). Companion to Figure 5.5's Plot 309770: that plot was chosen because it is unusually well-explained, and this one shows what the same attribution mechanism looks like for an ordinary plot instead.

Q2 appendix

Flagged-plot split diagnostics Around 10 plots recur as the worst residuals across every model and feature set (a flag rate roughly $36\times$ higher than chance, not a random tail). This traces to the fixed- k curve-fitting mechanism used to build $\Delta y_{\max}$ (Equation 4.7): growth rate is pinned to the plot's assigned yield class, so a plot that genuinely grows slower or less smoothly than that assumption forces an inflated ceiling estimate instead. Scaling this up with Tukey's $1.5 \times \text{IQR}$ fence flags 266 plots in the 4-survey cohort (0.47%) and 709 in the 6-survey cohort (5.32%), clustered by compartment. Removing these plots from training does not improve R^2 , they carry real signal, not just noise.

A population-wide split test (recomputed on 250 of these plots) asked whether each flagged plot's pattern looks like the fixed- k artefact or like genuinely fast growth: 71.6% fit the artefact signature (exposed terrain), and 28.4% look like genuinely fast, real growth (sheltered terrain), tested with actual p -values rather than judged by eye. Two open items are disclosed rather than resolved here: the 266-vs-250 count difference between the Tukey-fence count and the split-test sample has not been traced to the

exact plots involved, though both counts were independently verified; and the artefact mechanism is confounded with a real wind-exposure effect, since exposed terrain plausibly drives both genuinely unusual growth and the artefact pattern, and this data cannot fully separate the two. This is the same rate-ceiling-mixing mechanism described for the whole population in Section 4.6, just at a far more extreme scale for this flagged tail, not a second, unrelated finding.

Category-level Moran’s I diagnostics (with terrain bandwidth-artifact adjustment)

Q3 appendix

XGBoost hyperparameter search Grid: $n_estimators \in \{300, 500, 800\}$, $max_depth \in \{3, 4, 6\}$, $learning_rate \in \{0.02, 0.04, 0.08\}$, 27 configurations, each fit on Set3 with 5-fold spatial-block cross-validation (135 fits per cohort, 270 total across both cohorts). Model selection used mean validation R^2 only, never test R^2 , to avoid tuning-on-test leakage; test metrics are reported only for the winning configuration, after selection was already decided.

Cohort	n_{est}	Depth	LR	Val. R^2	Test R^2	Test RMSE	Test MAE
4-survey (winning)	500	6	0.02	0.682	0.674 ± 0.009	4.548 ± 0.137	3.431 ± 0.123
4-survey (borrowed default)	500	4	0.04	0.678	0.675 ± 0.009	4.539 ± 0.152	3.431 ± 0.153
6-survey (winning)	300	3	0.08	0.717	0.722 ± 0.031	3.656 ± 0.232	2.669 ± 0.183
6-survey (borrowed default)	500	4	0.04	0.712	0.718 ± 0.025	3.682 ± 0.221	2.697 ± 0.170

Table C.3: Winning configuration per cohort (selected on mean validation R^2 across 5 folds) against the generic default config borrowed from the same fitting code before this search was run. The 4-survey winning-by-validation config is marginally *lower* on test R^2 than the borrowed default (0.674 vs. 0.675), expected noise from selecting on validation rather than test, not a sign the search failed. Across both cohorts, the full 270-fit search moves test R^2 by at most 0.004, well inside the fold-to-fold SD (0.009 on 4-survey, 0.025–0.031 on 6-survey): the borrowed default was already close to optimal, and XGBoost’s accuracy edge over the neural models is not an artefact of a lucky or unfair hyperparameter choice.

Full 6-survey table

Model	RMSE↓	MAE↓	R^2 ↑
Linear	4.877±0.643	3.619±0.426	0.509±0.067
RF	4.062±0.249	3.010±0.185	0.656±0.046
XGBoost (tuned)	3.656±0.232	2.669±0.183	0.722±0.031
DNN	3.903±0.316	2.867±0.275	0.684±0.031
PINN	,	,	,
PINN- k	,	,	,

Table C.4: Model comparison for top-height prediction, 6-survey, Set3, spatial_block_kfold, per-fold mean $\pm$ SD across five folds. $n=82,614$ row-years (13,769 plots). Referenced from the main text's DNN-vs.-XGBoost fold-count and compartment-subsampling discussion. PINN/PINN- k were not rerun with the corrected forward pass on this cohort (–); see main-text Table 5.3's caption.

Q3 results tables

Fold	Split membership	Mean y_{max} deviation	Rows > 70 m (of 1484)
0	100% test	+12.1 m	72
1	95% train	+11.5 m	44
2	98% train	+10.7 m	0
3	100% train	+15.6 m	0
4	100% validation	+12.4 m	0

Table C.5: Compartment 1129's mean y_{max} deviation in each of the 5 spatial-block folds, by how much of the compartment's own data that fold trained on. Inflation persists (and does not shrink) even when the model trains on the compartment's own real data.

Fold	Plain PINN trunk mean ·	PINN- k trunk mean ·	Ratio
0	0.207	0.410	1.98×
1	0.238	0.553	2.33×
2	0.255	0.844	3.31×
3	0.339	0.668	1.97×
4	0.268	0.704	2.63×

Table C.6: Mean absolute size of the shared correction network's output, plain PINN vs. PINN- k , all 5 folds. PINN- k 's correction is larger than plain PINN's in every fold, roughly 2× or more throughout, not a strict trend.

Age band (years)	<i>n</i>	DNN MAE	PINN MAE	PINN- <i>k</i> MAE
15–30	29,837	2.49	2.52	2.53
30–45	17,861	2.69	2.85	2.93
45–60	7,590	4.16	4.24	4.42
60–75	2,034	4.63	5.23	5.15
75–93	751	6.83	6.40	6.81
Height band	<i>n</i>	DNN MAE	PINN MAE	PINN- <i>k</i> MAE
0–10 m	12,373	2.68	3.37	3.15
10–20 m	29,629	2.59	2.62	2.74
20–30 m	13,733	3.36	2.98	3.15
30–47 m	2,338	5.26	5.71	5.72

Table C.7: Mean absolute error by tree age and by tree height, DNN vs. plain PINN vs. PINN-*k*, pooled across all 5 folds (58,073 plots for all three models). Bold: DNN is the WORST of the three at the oldest ages, not the best, see note below on how this reverses the fold-0-only reading.

Compartment	Direction	Mean dev. (m)	Most extreme plot (m)	Why flagged
1129	Over	+12.2 (rank 10/178)	+25.8 (rank 1/178)	Single most extreme plot
2021	Under	-21.4 (rank 1/178)	-30.4 (rank 1/178)	Largest mean <i>and</i> most extreme plot

Table C.8: The two flagship compartments, ranked among the 178 compartments with $n \geq 50$ pooled held-out plots. 1129 wins on single-plot extremity alone; 2021 wins on both mean and single-plot extremity.

Compartment	Mean dev. (m)	<i>n</i> plots	Yield class	Agrees?	Mean age (yr)
<i>Over-productive, top 5 by mean deviation ($n \geq 50$)</i>					
2057	+15.6	112	10–16 (mixed)	No	59.8
1130	+14.6	78	22–24	Yes	33.6
2130	+14.6	98	20–24	Yes	35.1
1122	+13.4	51	24	Yes	25.0
1027	+13.4	56	12	No	72.0
<i>Under-productive, top 5 by mean deviation ($n \geq 50$)</i>					
2021	-21.4	403	12	Yes	,
1070	-18.1	222	12	Yes	,
2094	-17.7	279	10–12	Yes	,
1069	-11.8	297	12	Yes	,
2022	-10.3	417	16	Weakly	,

Table C.9: Top 5 over- and under-productive compartments by mean deviation ($n \geq 50$), checked against independent yield class. Reported as-is, not filtered down to the rows that agree. Under-productive side agrees with independent yield class in 4/5 cases; over-productive side agrees in only 3/5, including a disagreement from the single biggest over-productive signal (2057). The two disagreeing over-productive rows (2057, 1027) are both old and wind-exposed (mean age 59.8/72.0yr, windward index +15.2/+11.0); the three agreeing rows (1130, 2130, 1122) are both young and sheltered-to-neutral (mean age 25.0–35.1yr, windward index -9.5 to +0.4), see main-text discussion of this split.

Comparison	Pearson <i>r</i>	<i>n</i>
Plain PINN y_{max} deviation vs. yield class	0.273	58,073
PINN- <i>k</i> 's <i>k</i> vs. yield class	0.286	58,073
PINN- <i>k</i> 's y_{max} vs. yield class	0.135	58,073

Table C.10: Correlation between each model's personalisation and the Forestry Commission's independent yield-class rating, never a model input, checked against both feature lists. Pooled across all 5 folds.

Q3 supporting figures

PINN- k 's per-plot parameters, pooled across all 5 folds ($n=58,073$ plots): $y_{max,i}$ stays close to the population value; k_i shows a population-wide shift, not symmetric heterogeneity

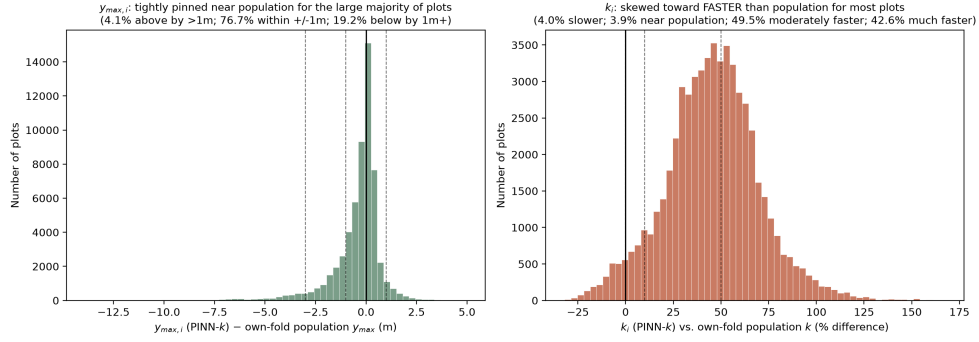


Figure C.3: PINN- k 's per-plot parameters, pooled across all 5 folds (58,073 plots), each plot's deviation measured against its own fold's population baseline. Left: $y_{max,i}$ stays tightly pinned near the population value for the large majority of plots (76.7% within $\pm 1m$). Right: k_i is skewed strongly toward faster than population for most plots (92.1% moderately or much faster), a population-wide shift, not a symmetric spread of real heterogeneity. Quantitative companion to Figure C.7's shape: the same one-sided k routing, shown as a distribution rather than a scatter.

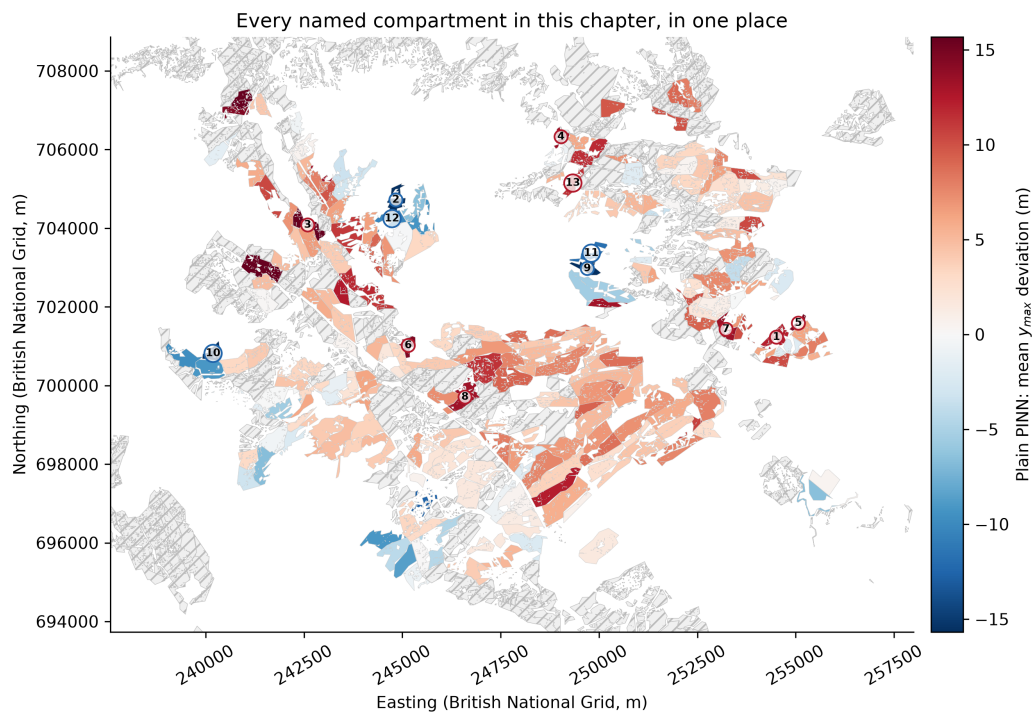


Figure C.4: Every compartment named anywhere in this chapter, numbered on one map, with the reason each was highlighted. A single lookup source, some compartments here appear in a table but no other figure, or a figure but no table.

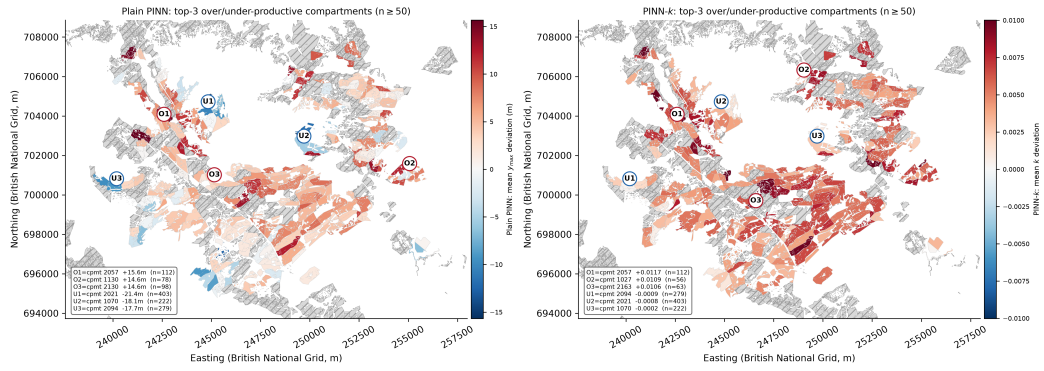


Figure C.5: Top-3 over- and under-productive compartments ($n \geq 50$), plain PINN (left) vs. PINN- k (right). 2057 tops both; the under-productive top-3 is the same three compartments on both sides, just reordered.

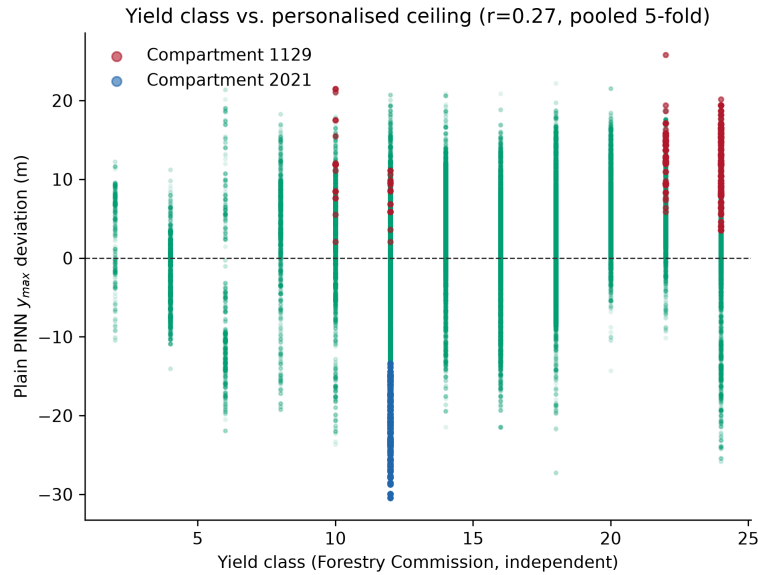


Figure C.6: Plain PINN's y_{max} deviation vs. independent yield class, pooled across all 5 folds. Green: every held-out test plot (58,073), one dot each, yield class only takes about a dozen discrete values, which is why the dots stack into vertical bands rather than spreading evenly across the x-axis. Red and blue: the same kind of point, restricted to the two flagship compartments (1129, 2021), so their position within the wider pattern is visible, 1129 clustered high (mostly yield class 22-24), 2021 clustered low and tightly (entirely yield class 12, strongly negative). The dashed black line is a zero-deviation reference, not a data series. Real but modest correlation (Table C.10), the upward drift of the whole cloud left to right is what that correlation number is measuring.

Why the broadest feature list looked worse Why the broadest environment list (Set4) seemed to underperform the medium list (Set3): its extra features are not ignored

by the model, they draw more than their fair share of permutation importance, they just do not repay that attention in held-out accuracy as well as the medium list's more curated features. Caveat: the Set3-vs-Set4 gap this explains is itself within fold-to-fold noise for plain PINN (Table 5.4), and absent entirely for PINN- k , a plausible explanation for a small, uncertain effect, not a solid one.

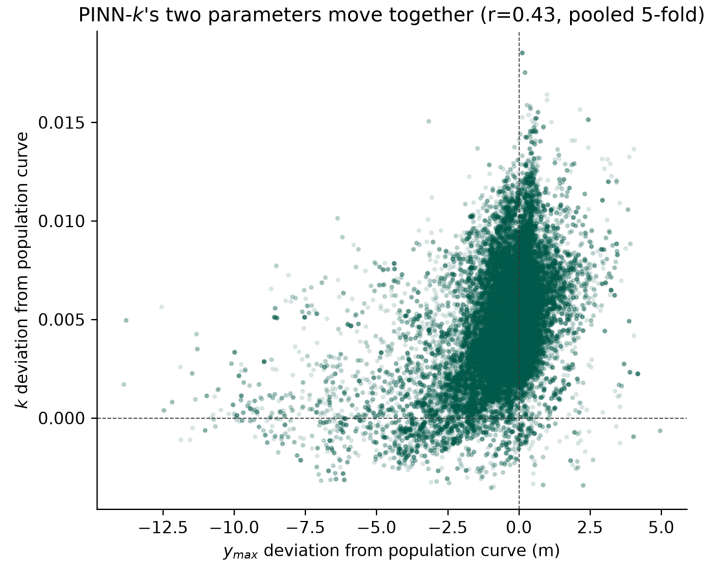


Figure C.7: Covariance between personalised parameters in PINN- k (pooled across 5 folds). Plot-level asymptote departures ($\Delta y_{\max}$) plotted against growth-rate departures (Δk) relative to each fold's regional baseline ($r = 0.43$). The positive relationship demonstrates coordinated biological scaling across diverse stand conditions.